Report

Emergent Particles and Gauge Fields in Quantum Matter

- Ben J. Powell

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Contents

1. Introduction

In this report, I will discuss Ben J. Powell's article titled *Emergent Particles and Gauge Fields in Quantum Matter* [1]. The article is meant to be a pedagogical overview of the fundamental aspects and ideas involved in contemporary research concerning correlated matter. The author starts from basic ideas of Adiabatic Continuity and Landau Theory, and goes through Nambu-Goldstone Bosons, Emergent Gauge fields, Anderson-Higgs Mechanism, eventually ending with the discussing of how topology is consequential in understanding certain states of matter.

I will be attempting to follow the references in the article to add relevant details wherever necessary.

2. Basic Goals

There are several parallels between High Energy/Particle Physics, and modern Condensed Matter Physics. Apart from shared methodologies, they also share a common goal - that of trying to understand how theories of nature unify at increasing energies, and symmetries in the theory "break down" at lower energies. Most research in the so-called Low-Energy Physics is dedicated to understanding the latter problem - the emergence of new phases of matter, and the broken symmetries associated with them.

Alongside such broken symmetries, new emergent particles are often found in low temperature phases of matter. A common example of this are phonons in crystals. Crystals break the continuous translation and rotation symmetries of the liquid phase, and have a certain rigidity to deformations that attempt to disrupt their spatial order. This rigidity is intimately connected with the arising of new "massless" particles called phonons, which are quantized lattice vibrations.

It has also become increasingly obvious that not all phase transitions correspond with a broken symmetry. In materials that display macroscopic quantum properties, the wavefunction that describes the material can undergo qualitative changes at a phase transition without breaking a symmetry. In such matter, one encounters exotic particles and new gauge fields, demonstrating properties that are not found in the standard particles like electrons, protons, etc.

3. The Standard Model of materials

To understand the low-energy excitations in materials, we need to scaffold our understanding using the so-called **Standard model of materials**. It consists of the following components:

1. Quasiparticles
2. Nambu-Goldstone Bosons
3. Anderson-Higgs Mechanism
4. Topological defects

3.1. Adiabatic Continuity and Quasiparticles

Modern Physics has knowledge of a "theory of everything" for most materials, i.e. it knows the microscopic laws that need to be evaluated to understand the behaviour of all materials. The problem lies in solving these equations exactly for macroscopic systems, which is most often impossible. There are two approaches left:

- Make severe aproximations: First principles approaches like Density Functional Theory are used in several contexts and can supply useful results.
- Study a simpler problem that captures the essential physics of the phenomena.

We will be focusing on the latter approach. Supposing that we have such a model that we can solve exactly (or very accurately), we need to be able to determine what predictions from the model are applicable to the real materials we are interested in. This is where the idea of **adiabatic continuity** is useful. The following example is instructional in explaining this concept.

3.1.1. 1 Dimensional Simple Harmonic Oscillator with a Dirac Delta perturbation

We begin with the Hamiltonian,

$$H = H_0 + H'$$

Where we have the unperturbed Hamiltonian

$$H_0 = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{1}{2}m\omega^2 x^2 \tag{1}$$

And the perturbation is given by

$$H' = \lambda\delta(x) \tag{2}$$

Where λ controls the strength of the perturbation. For $\lambda = 0$, we have the familiar 1D QSHO, with the energy eigenstates labelled by the non-negative integers $n = 0, 1, 2, ...$, with values $E_n = \hbar\omega\left(n + \frac{1}{2}\right)$.

The good quantum number n is interpreted as the number of nodes in the wavefunction $\psi_n(x)$.

Taking a perturbative approach, we can see that the perturbing dirac delta can only affect the even parity states and not the odd parity ones. This is due to the fact that the odd parity states have nodes at origin and do not "see" the dirac delta, i.e. integrals such as

$$\int_{-\infty}^{\infty} \psi_n^*(x)\delta(x)\psi_{m(x)} dx = 0 \forall n, m = 1, 3, 5, ...$$

The interesting physics is in the even parity states. An exact solution can be derived for these states in terms of parabolic cylinder functions [2], [3] and proceeds as follows:

First, we write down the full time-independent Schrodinger equation for the problem,

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} + \frac{1}{2}m\omega^2 x^2\psi(x) + \lambda\delta(x)\psi(x) = E\psi(x) \tag{3}$$

For $x \neq 0$, we have the 1D QSHO differential equation,

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} + \frac{1}{2}m\omega^2 x^2\psi(x) = E\psi(x)$$

We rescale the length coordinate to $\xi = \alpha x$, where $\alpha = \sqrt{\frac{m\omega}{\hbar}}$ is the characteristic lengthscale. We are left with,

$$\frac{\mathrm{d}^2\psi}{\mathrm{d}\xi^2} + (2\varepsilon - \xi^2)\psi(\xi) = 0 \tag{4}$$

Where $\varepsilon = \frac{E}{\hbar\omega}$ is the rescaled energy. This is of the form of a standard differential equation, whose square-integrable solutions are related to special functions called *Parabolic Cylinder Functions*, $D_\nu(z)$. The standard differential equation that is satisfied by $D_\nu(z)$ is,

$$\frac{\mathrm{d}^2 D_\nu}{\mathrm{d}z^2} + \left(\nu + \frac{1}{2} - \frac{1}{4}z^2\right) D_\nu(z) = 0$$

To map our dimensionless Schrodinger equation Eq. 4 to this standard form, we set $z = \sqrt{2}\xi \Rightarrow \frac{\mathrm{d}^2}{\mathrm{d}\xi^2} = 2\frac{\mathrm{d}^2}{\mathrm{d}z^2}$.

This yields,

$$\frac{\mathrm{d}^2\psi}{\mathrm{d}z^2} + \left(\varepsilon - \frac{1}{4}z^2\right)\psi = 0$$

Comparing the equations, we can identify $\nu + \frac{1}{2} = \varepsilon \Rightarrow \nu = \frac{E}{\hbar\omega} - \frac{1}{2}$. The boundary conditions are satisfies as the function $D_\nu(z)$ behaves as $e^{-\frac{z^2}{2}}$ for large $|z|$. Since we require an even-parity solution, we frame the general solution as:

$$\psi(\xi) = A D_\nu\left(\sqrt{2}\ |\xi|\right) \tag{5}$$

A is a normalization constant. Now, we enforce the delta function boundary condition at origin - it forces a discontinuity in the first derivative of the wavefunction at origin. To find this, we integrate the full time-independent equation Eq. 3 over an infinitesimal interval $[-\delta, \delta]$ and then we take the limit $\delta \to 0$.

$$-\frac{\hbar^2}{2m}\int_{-\delta}^{\delta} \frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} dx + \frac{1}{2}\int_{-\delta}^{\delta} m\omega^2 x^2 \psi(x) dx + \lambda \int_{-\delta}^{\delta} \delta(x)\psi(x) dx = E\int_{-\delta}^{\delta} \psi(x) dx$$

In the limit we are taking, all the integrals over the continuous terms vanish. We are left with,

$$-\frac{\hbar^2}{2m}\left(\left.\frac{\mathrm{d}\psi}{\mathrm{d}x}\right|_{0^+} - \left.\frac{\mathrm{d}\psi}{\mathrm{d}x}\right|_{0^-}\right) + \lambda\psi(0) = 0$$

Using the fact that the wavefunction is even, we have $\psi'(0^+) = -\psi'(0^-)$. This leads us to,

$$\left.\frac{\mathrm{d}\psi}{\mathrm{d}x}\right|_{0^+} = \frac{m\lambda}{\hbar^2}\psi(0)$$

Converting to our dimensionless coordinate $z = \sqrt{2}\alpha x \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x} = \sqrt{2}\alpha\frac{\mathrm{d}}{\mathrm{d}z}$,

$$\sqrt{2}\alpha \left.\frac{\mathrm{d}\psi}{\mathrm{d}z}\right|_{z=0} = \frac{m\lambda}{\hbar^2}\psi(0) \tag{6}$$

We evaluate the parabolic cylinder function at $z = 0$, which is related to the Gamma function,

$$\psi(0) = AD_\nu(0) = A\frac{\sqrt{\pi}2^{\frac{\nu}{2}}}{\Gamma\left(\frac{1-\nu}{2}\right)}$$

And the first derivative is,

$$\left.\frac{\mathrm{d}\psi}{\mathrm{d}z}\right|_{z=0} = AD'_\nu(0) = -A\frac{\sqrt{\pi}2^{\frac{\nu+1}{2}}}{\Gamma\left(\frac{-\nu}{2}\right)}$$

Plugging there identities into Eq. 6, and rearranging we arrive at an exact quantization condition:

$$\frac{\Gamma\left(\frac{1-\nu}{2}\right)}{\Gamma\left(-\frac{\nu}{2}\right)} = -\frac{m\lambda}{2\alpha\hbar^2}$$

We substitute back α and ν, to get an expression in terms of the energy,

$$\frac{\Gamma\left(\frac{3}{4} - \frac{E}{2\hbar\omega}\right)}{\Gamma\left(\frac{1}{4} - \frac{E}{2\hbar\omega}\right)} = -\frac{\lambda}{2\hbar\omega}\sqrt{\frac{m\omega}{\hbar}} \tag{7}$$

This is a transcendental equation for the allowed energies E, and it can not be solved in general for a closed form for a general λ.

The motivation for solving this problem is to demonstrate that this perturbation H' does not change *anything qualitative* about the solutions of the simpler problem of 1D QSHO. Note the following limits:

- Weak Coupling ($\lambda \to 0$): Here, the RHS of Eq. 7 approaches 0. For this to happen, the denominator on the LHS must diverge. We know that $\Gamma(z)$ has poles at $z = 0, -1, -2, \dots$. We set the argument of the Gamma function in the denominator to $-k \in \mathbb{Z}$, and we find

$$E = \hbar\omega\left(2k + \frac{1}{2}\right) \tag{8}$$

  Which are precisely the energy levels of the *unperturbed* even states of the 1D QSHO.

- Strong Coupling ($\lambda \to \infty$): Here, the RHS of Eq. 7 approaches $-\infty$. For this to happen, the Gamma function in the numerator must have a pole. Again, setting the argument of the numerator to be $-k \in \mathbb{Z}$, we find

$$E = \hbar\omega\left((2k+1) + \frac{1}{2}\right) \tag{9}$$

  Which are the energy levels of the *unperturbed* odd states of the 1D QSHO.

From this, two features of the model remain regardless of the value of λ:
1. The eigenstates labels remain the same, and still identify the number of nodes in the wavefunction as it is only in the strong coupling limit where the delta potential forces a node at the origin.
2. $E_n < E_m \Leftrightarrow n < m$, and in the strong coupling limit the even states become degenerate with the odd state above them but do not exceed them.

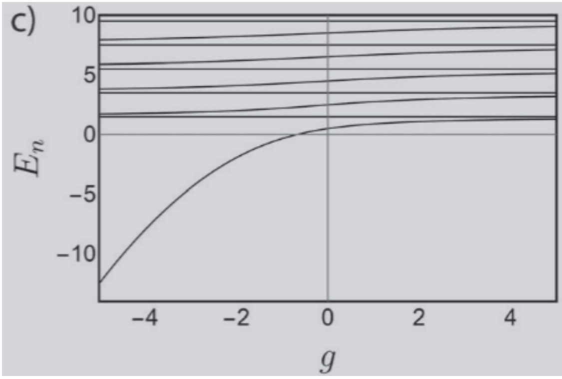

Figure 1: Plot of the energies of the first 10 levels of the delta-perturbed QSHO. Here, $\lambda \to g$. It is clear that the levels never cross. Source: [1]

The perturbation *does not change anything important* about the 1D QSHO solution, and this allows us to apply the predictions from that simple model to this one with the Dirac Delta potential at origin. This is the basic idea of adiabatic continuity, and any two problems that are connected by a smooth deformation of their Hamiltonians such that there are no level crossings (and hence no avoided crossings) are said to be adiabatically continuous/connected. This can also be stated more precisely: Hamiltonians are said to be adiabatically connected if the set of quantum numbers does not change, and the ordering of the energy eigenvalues does not change when one Hamiltonian is slowly transformed into the other [1].

However, in several interesting systems even arbitrarily small perturbations can change the essential physics. When we are dealing with such a system in the thermodynamic limit, such a failure of adiabatic continuity indicates a phase transition. This is important, as phase transitions are easy to detect (theoretically and experimentally), and thus establishing adiabatic continuity between a toy model and a real system tells us that both share the same thermodynamic phase. This, in turn, means that they will share their most important characteristics as well.

3.1.2. Landau Fermi Liquid Theory

Simple estimates tell us that in simple metals such as gold, copper, etc. the total potential energy due to Coulombic interactions between electrons is much larger than kinetic energy of the electrons at the Fermi surface. Yet, theories such as Drude Theory and the more sophisticated Sommerfeld Free Electron theory can make quantitative (correct) predictions for many properties of such metals. In fact, to explain most of the properties of these metals only an understanding of weakly interacting fermions is required. Landau's Fermi Liquid Theory shows this, by starting with a gas of free (i.e. non-interacting) fermions and slowly turning on the interactions between them. It turns out that the interacting and non-interacting Hamiltonians are adiabatically connected.

Low-energy excitations in non-interacting gases of fermions involve taking single particle from below the Fermi energy and exciting it to above the Fermi energy. This could be framed as the excited state containing one particle above the Fermi energy, and a *hole* in the otherwise filled sea of fermions below the Fermi energy. Both of these are characterized entirely by their spin and momentum.

In Fermi Liquid theory, the fundamental low-energy excitations are also fermions and holes - but they can no longer be thought of as single particle excitations. This is because such an excitation now involves changes in the quantum states of several electrons - it is a many-body state. Regardless of this change in identity, the excitations are guaranteed to share the same quantum numbers as the single particle states of the non-interacting system due to adiabatic continuity. Therefore, these excitations in the Fermi Liquid are called **Quasiparticles**.

There are a few differences between the low-energy excitations in the non-interacting case and the quasi-particle excitations in a Fermi Liquid:

- The Particle-hole excitations are exact energy eigenstates of the non-interacting Fermi gas. As a result, they are stable and do not decay. On the other hand, quasipartices are *not* exact energy eigenstates of the interacting Fermi Liquid - they interact with each other and thus have a finite lifetime.
- The quasiparticle can be essentially imagined as an electron "dragging" a cloud of particle-hole excitations as a screening cloud along with it. This "drag" effectively increases the mass of the quasiparticle as compared to the bare electron.

A detailed introduction to Landau's Fermi Liquid Theory is beyond the scope of this report, and can be found in [4] and [5].

3.1.3. Adiabatic continuity is not guaranteed

Not all systems that consist of interacting fermions are Fermi Liquids, and various instabilities can arise that lead to other phases of matter. These could have strongly interacting fermions, leading to low-energy excitations that are not the single-particle like excitations of the Fermi Liquid.

There are examples of states of matter, such as *strange metals* where quasiparticles are not found. In 1D metals, one encounters the Luttinger Liquid phase, where there are two distinct quasiparticles called spinons and holons that carry spin and charge respectively.

3.2. Spontaneously Broken symmetry

3.2.1. Landau theory of ferromagnetism

Ferromagnetism is defined as the long range order of the magnetic moments of atoms in a material. Again, Landau theory allows us to bypass the microscopic details of the system and understand quite a bit about the phase via an analysis based entirely on the symmetries of the system. We define a coarse-grained *order parameter* - which is defined to be zero in the high-temperature disordered (paramagnetic) phase, and is non-zero in the ordered (ferromagnetic) phase. In this case, the order parameter is simply the magnetization $\boldsymbol{m}$.

Near the critical point of the ferromagnet-paramagnet transition, $\boldsymbol{m}$ is small and the free energy difference between the two phases can be expressed as a power series in $\boldsymbol{m}$:

$$\Delta F = \alpha|\boldsymbol{m}|^2 + \beta|\boldsymbol{m}|^4 + \ldots \tag{10}$$

Note that we have to respect the symmetries of the system while writing this power series, namely the fact that most systems that display ferromagnetism have no preferred direction for magnetisation and are invariant under the transformation $\boldsymbol{m} \to -\boldsymbol{m}$. As a result, we can not have odd terms in the series. We can choose to truncate $\mathcal{O}(|\boldsymbol{m}|^3)$, and α and β remain as parameters of the theory.

Note that we require $\beta > 0$, otherwise $\boldsymbol{m} \to \infty$ and $\boldsymbol{m} \to -\infty$ would become the stable states of the system, which is not physical. We can put in our temperature dependence in α by declaring,

$$\alpha = \alpha'(T - T_c) \tag{11}$$

Where α' is some constant, and T_c is the critical temperature of transition. So, we are left with the expression:

$$\Delta F = \alpha'(T - T_c)|\boldsymbol{m}|^2 + \beta|\boldsymbol{m}|^4 \tag{12}$$

If we represent this with ΔF along the z-axis and $\boldsymbol{m}$ lying on the xy-plane, we would get the famous mexican hat potential energy landscape for $T < T_c$. In the high temperature state of $T > T_c$, there is only one stable value of ΔF, which is zero, corresponding to $\boldsymbol{m} = 0$. In contrast, in the low temperature state of $T < T_c$, there is no stable value of ΔF at $\boldsymbol{m} = 0$ and the system chooses to lie along a circle of radius $|\boldsymbol{m}| = \sqrt{\frac{|\alpha|}{2\beta}}$ lying on the xy-plane. This is the ferromagnetic phase.

But, the bottom of the mexican hat has continuous $U(1)$ symmetry, and the system has to choose a certain value to settle into to minimize free energy. In reality, the choice of direction is made by the various stray magnetic fields that couple to the system $\boldsymbol{h}$, adding a term $-\boldsymbol{h} \cdot \boldsymbol{m}$ to the free energy expression. This breaks the continuous rotation symmetry of the system, and selects a direction for the magnetization.

The low energy $\boldsymbol{m} \neq \boldsymbol{0}$ state has thus *broken a symmetry of the underlying Hamiltonian* in the low-temperature phase. This is called **spontaneously broken symmetry**.

This theory is clearly classical in nature, so we proceed to discuss how it should be adapted to deal with quantum systems.

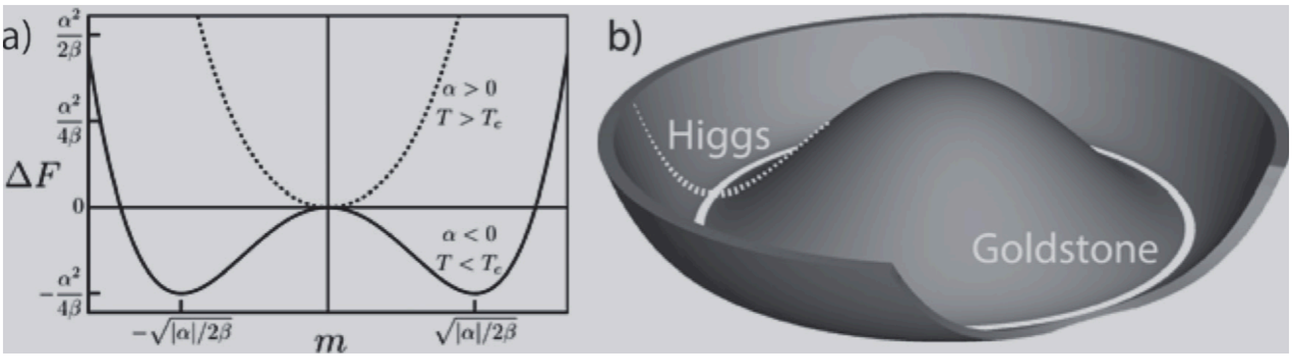

Figure 2: The famous Mexican hat potential energy landscape, characteristic of the low-temperature ferromagnetic phase. Source: [1]

3.2.2. Spontaneously broken symmetries in Quantum mechanics

The magnetisation of a ferromagnet can be easilt connected to the microscopic quantum behaviour of, say, N electronic spins:

$$\boldsymbol{m} = \langle \widehat{\boldsymbol{m}} \rangle = \frac{1}{N} \left\langle \sum_{i=1}^{N} \hat{\boldsymbol{S}}_{\boldsymbol{i}} \right\rangle \tag{13}$$

Where $\hat{\boldsymbol{S}}_{\boldsymbol{i}} = (S_i^x, S_i^y, S_i^z)$ is the vector spin operator of the i-th electron. For most simple systems, the Hamiltonian commutes with either the total spin (in the case of interacting spins) or just a component of the spin (in case of coulombic interactions) - most often chosen to be the z-component for conventional reasons. This allows for the spin eigenvalues to be good quantum numbers for the problem at hand. With this information, we can now investigate how spontaneously broken symmetry acts in such spin systems with the example of the two-site Heisenberg model.

3.2.3. Two Site Heisenberg Model

The Hamiltonian for this model is given as:

$$\hat{H} = J\hat{\boldsymbol{S}}_{\mathbf{1}} \cdot \hat{\boldsymbol{S}}_{\mathbf{2}} + h\sum_{i=1}^{2} S_i^z \tag{14}$$

Where the sign of J determines if the couplings are antiferromagnetic or ferromagnetic, and h is the external field. The total spin operator $\hat{\boldsymbol{S}} = \hat{\boldsymbol{S}}_{\mathbf{1}} + \hat{\boldsymbol{S}}_{\mathbf{2}}$ and the total z-component of spin $S^z = S_1^z + S_2^z$ commute with the Hamiltonian, making their eigenvalues the good quantum numbers for the states $|S, S^z\rangle$. In the two-spin basis, these states are (via the usual addition of angular momenta):

$$|0,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow,\downarrow\rangle - |\downarrow,\uparrow\rangle)$$

Which is called the spin-singlet state, and

$$|1,1\rangle = |\uparrow,\uparrow\rangle$$

$$|1,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow,\downarrow\rangle + |\downarrow,\uparrow\rangle)$$

$$|1,-1\rangle = |\downarrow,\downarrow\rangle$$

Called the spin-triplet states.

For ferromagnetic interactions ($J < 0$) and no field ($h = 0$), the triplet states form the threefold degenerate ground state of the system, and for the antiferromagnetic case ($J > 0$) with no field the singlet state forms the ground state.

Now, if we are introduce a small field h (like the case of Landau Theory's SSB) in the Ferromagnetic case, say some $h > 0$, then the $|1,-1\rangle$ state is selected as the ground state as the degeneracy is lifted. Turning off the field adiabatically $h \to 0$, the system remains in the $|1,-1\rangle$ state as this is an *exact eigenstate* for *all* values of h. Thus, the no-field Hamiltonian has full spin rotation symmetry, but the *prepared* ground state breaks the symmetry. The case is different for the antiferromagnetic case, as the ground state $|0,0\rangle$ retains the full spin rotation symmetry of the Hamiltonian. Small fields can not cause a level crossing (where $|1,-1\rangle$ becomes the ground state), and thus the symmetry of the Hamiltonian is preserved.

3.2.4. Neel Antiferromagnets

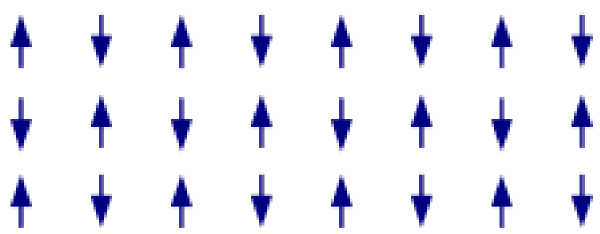

Figure 3: Antiferromagnetic order. Source: Wikipedia

The above discussion seems to suggest that antiferromagnets do not occur as spontaneously symmetry broken ground states - but this is in contradiction with experimental evidence.

The simplest kind of antiferromagnets have Neel order - the spins alternate in direction on the lattice, cancelling each other's magnetic moment. To describe this state, we need to account for this alternating spin direction in our order parameter. Hence, we define a "staggered" magnetisation,

$$\boldsymbol{m}_s = \langle \widehat{\boldsymbol{m}}_s \rangle = \frac{1}{N} \left\langle \sum_{i=1}^{N} (-1)^i \hat{\boldsymbol{S}}_{\boldsymbol{i}} \right\rangle \tag{15}$$

The symmetry broken Neel states of $|\uparrow, \downarrow\rangle$ or $|\downarrow, \uparrow\rangle$ are not eigenstates of the two-site Heisenberg problem Eq. 14, and on top of that a constant magnetic field will not pick these states out as the ground state. To adjust for that, we "stagger" our field the same way,

$$\hat{H} = J \hat{\boldsymbol{S}}_{\boldsymbol{1}} \cdot \hat{\boldsymbol{S}}_{\boldsymbol{2}} + h^S \sum_{i=1}^{2} (-1)^i S_i^z \tag{16}$$

Let us take the strong field limit, $\left(\frac{h^S}{J}\right) \to \infty$. The first term can be ignored, and it is obvious that the ground state is $|\uparrow, \downarrow\rangle$. Taking $h^S \to 0$ adiabatically, the system is not stable in this state - it returns to the rotationally symmetric $|0, 0\rangle$ state. In fact, this result is quite general, and our two site model is a special case of the more general Heisenberg model on a bipartitite lattice. A bipartite lattice is one where each lattice point belongs to one of two sublattices, $\mathcal{A}$ or $\mathcal{B}$. Interactions between points in the same sublattice are not allowed, and interactions between points in different sublattices is allowed. The following theorem holds for such lattices:

Theorem 3.2.4.1 (Marshall's Theorem): For any bipartite lattice with equal size sublattices, the ground state of the antiferromagnetic Heisenberg model on the lattice is the $S_{\text{tot}} = 0$ spin-singlet state.

The two-site case can be seen to be a bipartite lattice with $\mathcal{A} = \{1\}$ and $\mathcal{B} = \{2\}$, so Theorem 3.2.4.1 holds for it. This theorem seems to be saying that, in general, we can not have a broken symmetry ground state - as an antiferromagnetic bipartite lattice must have equal sized sublattices (one having up spins, and the other down spins).

The resolution to this problem can arrived at by noting that the staggered magnetisation operator (our order parameter for the antiferromagnetic state) is not part of the set of commuting operators for Eq. 14, i.e.

$$\left[\hat{H}, \hat{m}_S^z\right] \neq 0$$

This means that Neel states (the eigenstates of the staggered magnetisation operator) are *not stationary states*. We have already noted this in the strong field limit case for Eq. 16. Also, note that the singlet state is a *quantum superposition* of various Neel broken symmetry states. Such a superposition can be thought of as a quantum system rapidly (quantum) tunnelling between all the "classical" states that make up the superposition. Such a tunnelling has a characteristic timescale, which is entirely determined by the energy barrier between the states. The larger the barrier between the states, the slower the timescales over which a tunnelling event has a chance of occuring.

For small systems such as the two-site model, preparing a Neel state with a staggered fields and then taking $h^S \to 0$ adiabatically will allow the ground state to become the singlet state over very small timescales. This is due to the very small energy barrier between the states. But, it can be shown that the larger such a system gets, the higher the energy barrier is. In the thermodynamic limit of $N \to \infty$, the barrier becomes infinite and a tunnelling event can no longer occur. So, if a Neel state is prepared by a staggered field in a large enough system, the system in the Neel broken symmetry state "stays" in that state when the field is turned off - effectively breaking the symmetry of the Hamiltonian over the timescales that concern us. Mathematically, this can be precisely expressed as the non-commutativity of limits on the staggered magnetisation eigenvalue:

$$\lim_{h^S \to 0} \lim_{N \to \infty} m_S \neq 0$$

But,

$$\lim_{N \to \infty} \lim_{h^S \to 0} m_S = 0$$

Simple calculations indicate that to demonstrate this rate-limited broken symmetry the systems need not be *too* large - even with a few hundred spins a Neel state can have a lifetime on the order of years.

This demonstrates that phase transitions that involve order parameters that do not commute with the Hamiltonian can only happen in the Thermodynamic limit. In contrast, phase transitions where the order parameter does commute with the Hamiltonian, the thermodynamic limit is not *required* - we can see the paramagnetic to ferromagnetic transition in small systems such as oxygen molecules.

Real magnetic materials of course have many fine- and hyperfine-structure corrections which can explicitly break the symmetry of he Hamiltonian when added in, but this is *explicit* symmetry breaking and is distinct from the *spontaneous* symmetry breaking that we have discussed here.

3.3. Nambu-Goldstone Bosons

3.3.1. Superfluidity

The superfluid state is a many-body quantum phenomena where a macroscopic fluid in this state flows without any friction, i.e. superfluids have no viscosity. It is seen in several systems at very low temperatures, and when it occurs in electron fluids it is called *superconductivity*.

Ginzburg-Landau theory for superfluids does not have any microscopic justifications, it begins with the appropriately guessed macroscopic order parameter $\Psi(\boldsymbol{r})$ - a complex scalar field. Essentially, this is a macroscopic wavefunction, which reflects the fact that the superfluid seems to be one large macroscopic quantum state - an idea that was later justified using Microscopic theory.

Here, the two components of the complex scalar field physically represent the following:

- $|\Psi(\boldsymbol{r})|^2 = n_{S(\boldsymbol{r})}$, which is the local superfluid particle density. When it is non-zero, it denotes that a non-zero fraction of the material has entered the superfluid state.
- The phase $\theta(\boldsymbol{r})$ (in polar form, the order parameter is $\Psi(\boldsymbol{r}) = |\Psi(\boldsymbol{r})| e^{i\theta(\boldsymbol{r})}$) dictates the coherent macroscopic dynamics and transport properties of the superfluid. Applying the momentum operator $\hat{\boldsymbol{p}} = -i\hbar\boldsymbol{\nabla}$ to the wavefunction helps us identify the center of mass velocity of the superfluid flow:

$$\boldsymbol{v}_s = \frac{\hbar}{m}\boldsymbol{\nabla}\theta(\boldsymbol{r}) \tag{17}$$

Where m is the mass of the appropriate boson.

With these definitions in place, we can write down a free energy difference between the high- and low-temperature phases as a power series near the transition:

$$\Delta F = \alpha'(T - T_c)|\Psi|^2 + \beta|\Psi|^4 \tag{18}$$

Which is a $\phi - 4$ Landau theory like Eq. 12, because the free energy is *real and analytic* - meaning that it can not depend on odd powers of Ψ or $|\Psi|$. Moreover, the physical observables of a neutral superfluid must remain invariant under a *global* $U(1)$ transformation. This means that an arbitrary phase shift of $\Psi \to \Psi e^{i\theta}$ can not affect the free energy.

From Eq. 18 we can see that:

1. For $T > T_c$, the free energy is minimized when $\Psi = 0$. This means that the material is not in a superfluid state.
2. For $T < T_c$, the free energy is minimized by some $\Psi = \Psi_0 e^{i\theta}$, where $\Psi_0 = \sqrt{\frac{\alpha}{2\beta}}$. We have a freedom in θ, as any choice of it leads to the same free energy.

3.3.2. Phonons as Nambu-Goldstone Bosons

So far, the order parameter of the system has been treated as having a uniform value over the entire system. But, it can have spatial and temporal variations over the extent of the system, and this can have important consequences.

The Free Energy functional Eq. 18 is a thermodynamic scalar and only dictates the equilibrium configuration of the system. If we wish to analyze time-dependent phenomena like how the order parameter relaxes, how it supports propagating excitations (modes), and how it interacts with external forces, we need dynamical equations of motion. To do this, we start off with an effective field theory Lagrangian that respects the symmetries of the system:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \Psi^*)(\partial^\mu \Psi) - V(|\Psi|) \tag{19}$$

Where Einstein summation is indicated, and the first (kinetic) term is contracted with a metric $\eta = \mathrm{diag}(+1, -1, -1, -1)$, $\partial_\mu = \left(\frac{1}{c^*}\partial_t, \boldsymbol{\nabla}\right)$. Note that this is *pseudo-Lorentz invariant*, which the crucial distinction of $c \to c^*$ to reflect the fact that these excitations might not propagate at the speed of light. The potential V is simply the Landau Free Energy Eq. 18.

Let us work with the low-temperature broken symmetry field Ψ_0. We introduce a real scalar amplitude fluctuation field $\rho(\boldsymbol{r}, t)$ and a real phase fluctuation field $\theta(\boldsymbol{r}, t)$, leaving us with:

$$\Psi(\boldsymbol{r}, t) = (\Psi_0 + \rho(\boldsymbol{r}, t))e^{i\theta(\boldsymbol{r}, t)} \tag{20}$$

With this, the first term of Eq. 19 is,

$$\left(\partial_\mu \Psi^*\right)\left(\partial^\mu \Psi\right) = \left(\partial_\mu \rho\right)^2 + \left(\Psi_0 + \rho\right)^2 \left(\partial_\mu \theta\right)^2$$

Note that the energy penalty for local spatial variations in the order parameter is accounted for in this derivative - there is no need to put it in manually in the Landau Free Energy expression.

Next we expand the Landau potential term V upto quadratic order around Ψ_0, noting that the first derivative is zero by definition in the low-temperature phase, and the 2nd derivative evaluates to:

$$V''(\Psi_0) = -4\alpha = 4|\alpha| \equiv m_\rho^2$$

With the potential becoming,

$$V(\Psi_0 + \rho) \approx V(\Psi_0) + \frac{1}{2} m_\rho^2 \rho^2$$

Putting this into Eq. 19,

$$\mathcal{L}_{eff} \approx \frac{1}{2}\left(\partial_\mu \rho\right)^2 - \frac{1}{2} m_\rho^2 \rho^2 + \frac{1}{2}\Psi_0^2 \left(\partial_\mu \theta\right)^2 - V(\Psi_0) \tag{21}$$

The effective Lagrangian has thus neatly decoupled into two distinct real scalar field Lagrangians, each with fundamentally different properties.

- The Amplitude/Higgs Mode: The Lagrangian for the amplitude fluctuation field,

$$\mathcal{L}_\rho = \frac{1}{2}\left(\partial_\mu \rho\right)^2 - \frac{1}{2} m_\rho^2 \rho^2 \tag{22}$$

  is the Lagrangian for a *massive real scalar field.* This can be seen clearly, as the 2nd term looks like a potential term with an appropriate mass factor m_ρ. As a result, the variation in the amplitude of the order parameter Ψ even at zero momentum requires energy. The mass term represents the energy requires to "push" the system up the walls of the Mexican hat potential.

- The Phase/Nambu-Goldstone Mode: The Lagrangian for the phase fluctuation field,

$$\mathcal{L}_\theta = \frac{1}{2}\Psi_0^2 \left(\partial_\mu \theta\right)^2 = \frac{1}{2}\Psi_0^2 \left(\frac{1}{\left(c^*\right)^2}\left(\partial_t \theta\right)^2 - (\nabla \theta)^2\right) \tag{23}$$

  Represents a *massless real scalar field*, as there is no θ^2 term. **This is a direct manifestation of Goldstone's Theorem**: The spontaneous breaking of a continuous symmetry (The $U(1)$ symmetry in this case) mandates the existence of a massless (i.e. gapless) bosonic mode. There is no potential barrier to be overcome in rotating the system about the bottom of the Mexican hat, and the only energy costs are incurred when there is a spatial or temporal gradient in the phase. We will be focusing on this field, as we are interesting in *low-energy* excitations.

The Ψ_0 factor in Eq. 23 can be understood to be a measure fo the rigidity of the superfluid to the introduction of a phase difference between two spatially separate points in the material. Using the principle of least action (i.e. taking a functional derivative of the action or using the Euler Lagrange equations) on Eq. 23, we arrive at an equation of motion of the phase:

$$\nabla^2\theta = \frac{1}{c^*}\frac{\partial^2\theta}{\partial t^2} \tag{24}$$

Which is a typical wave equation, and has the solutions $\omega = c^* q$ or equivalently $E = c^* p$, where $E = \hbar\omega$ is the excitation energy, ω is the excitation frequency, $p = \hbar q$ is the momentum of the excitation, and q is the wavenumber. This is clearly a massless mode with linear dispersion, and it is called so in comparison to the relativistic dispersion relation $E^2 = m^2c^4 + p^2c^2$.

This masslessness also allows the for excitation energy E to go to zero when $\lambda \to \infty \Rightarrow q \to 0$, i.e. long wavelength and low-energy excitations - hence the name *gapless*. c^* is the speed of propagation of this excitation, and we call it the *phonon*. The quantization of a classical mode *usually* results in recovering bosonic statistics, and thus these are called *Nambu-Goldstone Bosons*.

3.3.3. Counting Nambu-Goldstone Bosons

Some phase transitions result in the breaking of multiple (continuous) symmetries, not just one like the previous case of the superfluid. In the case of an antiferromagnet, 2 symmetries are broken - rotation of the spin about the x and y axes, due to the selection of a particular direction for the spins. Again, a Lagrangian for the low-energy excitations, respecting the symmetries of the system can be written down:

$$\mathcal{L}_{NL\sigma M} = \frac{\rho_{AFM}}{2}\left[\frac{1}{(c^*)^2}\dot{\boldsymbol{n}}_s{}^2 - (\boldsymbol{\nabla}\cdot\boldsymbol{n}_s)^2\right] \tag{25}$$

Which is called a *Non-Linear Sigma Model* [1]. ρ_{AFM} is called the "spin-wave stiffness", and $m_s\boldsymbol{n}_s = \boldsymbol{m}_s$ with $|\boldsymbol{n}_s| = 1$ and $m_s = |\boldsymbol{m}_s|$. Plugging this Lagrangian into the Principle of Least Action yields two *independent* wave equations,

$$\nabla^2 n_s^x = \frac{1}{(c^*)^2}\frac{\partial^2 n_s^x}{\partial t^2}$$

and,

$$\nabla^2 n_s^y = \frac{1}{(c^*)^2}\frac{\partial^2 n_s^y}{\partial t^2}$$

Clearly, here we have two indpendent low-energy excitations represented by two Nambu-Goldtone modes. These excitations are called *magnons*, and they are massless with linear dispersion.

Similarly, crystallization in 3 dimensions breaks 3 continuous translation symmetries along the x, y, and z directions. A similar analysis as the Non-linear sigma model leads us to find 3 low-energy Nambu-Goldstone modes in crystals, representing one longitudinal and two transverse phonons.

This idea represents a more general fact of **Goldstone's Theorem** - If the low energy dynamics is governed by a pseudo-Lorentz invariant Lagrangian, then the number of Nambu-Goldstone modes is equal to the number of broken symmetries in that phase.

3.3.4. Absence of Pseudo-Lorentz invariance

Let us now look at the case of the Ferromagnet. Here, the two continuous rotation symmetries that are broken in the antiferromagnet are also broken, and so we would expect two Nambu-Goldstone modes with linear dispersion in each for the low-energy excitations.

Now, for the case of the Ferromagnet the effective low-energy Lagrangian is *not pseudo-Lorentz invariant*. Essentially, this is due to the nature of the commutation relations of the spin operators. Regardless, the low energy Lagrangian for ferromagnets is:

$$\mathcal{L}_{FM} = m\frac{n^y\dot{n}^x - n^x\dot{n}^y}{1+n^z} + \frac{\rho_{FM}}{2}\left[\frac{1}{(c^*)^2}\dot{\boldsymbol{n}}^2 - (\boldsymbol{\nabla}\cdot\boldsymbol{n})^2\right] \tag{26}$$

Where the additional first term is called the *Wess-Zumino Term*. The addition of this term can be motivated in two ways:

- It is required in order for the quantum theory to reduce to the appropriate Landau-Lifshtiz equations in the classical limit.
- The term is required to account for the Berry phase accumulated by the precessing spins.

A detailed discussion of this term is well beyond the scope of this report, and I will be describing the effects it has on the dynamics of the low-energy excitations.

Using Eq. 26 and the Principle of Least Action, and the additional assumption that $1 - n^z$ is small, we arrive that the linearized Landau-Lifshitz equatons.

$$\dot{n^x} = \frac{\rho_{FM}}{2}\nabla^2 n^y$$

and,

$$\dot{n^y} = -\frac{\rho_{FM}}{2}\nabla^2 n^x$$

Note that these are *coupled differential equations*. The solution is:

$$\omega = \frac{\rho_{FM}}{2}p^2$$

This is a *quadratic dispersion relation*, and not a linear one as we have previously found. Note that in the case of the antiferromagnet, this is avoided, as the *components* of the staggered magnetisation (which is our order parameter for antiferromagnets) along the x- and y- directions commute and are thus decoupled from each other. This can be seen from:

$$[\hat{m}_s^x, \hat{m}_s^y] = i\hbar\hat{m}^z$$

and,

$$m^z \equiv \langle\hat{m}^z\rangle = 0$$

With the latter being true, because the low-temperature antiferromagnetic phase can not have a net magnetisation along the z-dirextion by definition. This decoupling means that here is no Wess-Zumino like term in the corresponding low-energy effective theory for the antiferromagnetic phase - the non-linear sigma model is the correct description.

This example is a specific case of a more general principle: Whenever the effective Lagrangian has a pair of non-commuting order parameters, there is a *topological* term in the Lagrangian involving both the variables. This interaction terms results in the two decoupled linear dispersion Nambu-Goldstone modes being replaced with a single *quadratically dispersion* Nambu-Goldstone mode.

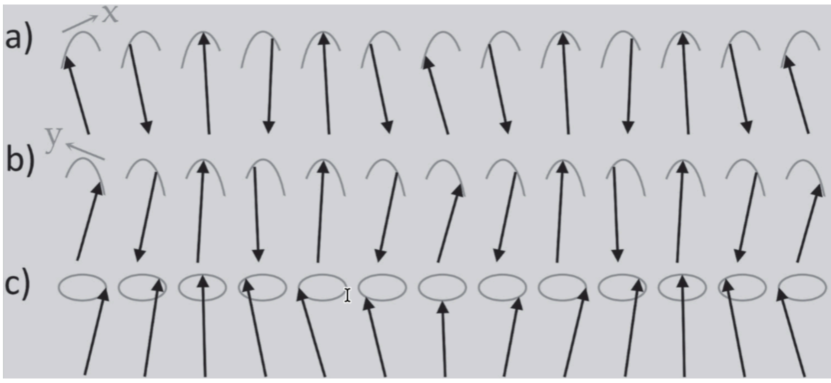

Figure 4: An illustration of the difference between magnons in antiferro- and ferromagnets. (a) and (b) represent the two transverse linear dispersion magnons in antiferromagnets. (c) shows the quadratically dispersion magnon in ferromagnets due to the Wess-Zumino term. Source: [1]

3.4. Massive excitations

Before we begin the discussion regarding excitations in matter that appear to have "mass", we must first discuss the idea of gauge invariance.

3.4.1. Gauge invariance in electromagnetism

In classical electromagnetism, the physical observables (in terms of which the Maxwell equations are usually stated) are:

- The electric field $\boldsymbol{E}$
- The magnetic field $\boldsymbol{B}$

To solve the Maxwell equations, it is frequently convenient to introduce the scalar potential ϕ and vector potential $\boldsymbol{A}$, relating to the physically observable fields as:

$$\boldsymbol{E} = -\boldsymbol{\nabla}\phi - \frac{\partial \boldsymbol{A}}{\partial t}$$

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}$$

In (covariant) relativistic notation, these potential fields can be packaged into a 4-potential,

$$A_\mu = (\phi, -\boldsymbol{A})$$

And the fields are encoded into the anti-symmetric field strength tensor:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{27}$$

But the potential A_μ does not *uniquely* determine the fields $\boldsymbol{E}$ and $\boldsymbol{B}$. If we apply a *local* transformation to the potentials using some arbitrary $\Lambda(x^\mu)$:

$$A_\mu \to A'_\mu = A_\mu + \partial_\mu \Lambda \tag{28}$$

The field strength tensor $F_{\mu\nu}$ remains unchanged, and hence the observable fields also remain unchanged. The transformation Eq. 28 is called a **gauge transformation**, and the fact that the physics does not change in this transformation is called **gauge invariance**. The potentials A_μ thus contains some redundancy in the description of the physics.

3.4.2. Local Phase Invariance in Quantum mechanics

Let us begin with Dirac's relativistic Lagrangian for a free fermion:

$$\mathcal{L} = \overline{\psi}(i\gamma_\mu \partial_\mu - m)\psi \tag{29}$$

This is expressed in natural units ($\hbar = c = 1$), with ψ being the Dirac spinor for the fermion, m its mass, and γ_μ being the 4×4 Dirac matrices. Eq. 29 has a *global* $U(1)$ symmetry - this means that under the transformation:

$$\psi \to e^{iq\alpha}\psi$$

for some constant α keeps Eq. 29 invariant. q is the conserved charge due to the $U(1)$ symmetry, as ensured by Noether's Theorem. In this case, it is the electric charge of the Fermion.

However, the principles of relativity prohibits us from changing the phase of the wavefunction ψ uniformly over all space-time simultaneously, as information can not trave faster than light. Such an agreement with special relativity is crucial here, as the Dirac equation is constructed to reflect the relativistic dispersion relation $E^2 = p^2 + m^2$. To reflect this fact, we must manually change our descriptions of this system to account for a *local gauge symmetry* under the transformation:

$$\psi(x) \to \psi'(x) = e^{iq\Lambda(x)}\psi(x) \tag{30}$$

Putting this transformed wavefunction into Eq. 29 does *not keep it invariant*, as the standard derivative acting on it yields an extra term:

$$\partial_\mu \psi' = e^{iq\Lambda(x)}\big(\partial_\mu \psi + iq(\partial_\mu \Lambda)\psi\big)$$

To restore/establish symmetry under such a transformation, we replace our usual derivative ∂_μ with a *covariant derivative* D_μ that is designed to transform under Eq. 30 as:

$$D'_\mu \psi' = e^{iq\Lambda(x)} D_\mu \psi \tag{31}$$

To achieve such a derivative, we introduce a new vector field A_μ and define the covariant derivative as:

$$D_\mu \equiv \partial_\mu - iqA_\mu$$

Additionally, for Eq. 31 to hold we require this field to transform as:

$$A_\mu \to A'_\mu = A_\mu + \partial_\mu \Lambda(x)$$

Note that this is *exactly the same* as Eq. 28, the gauge transformation of the components of the classical electromagnetic 4-potential. The locally invariant Lagrangian thus becomes,

$$\mathcal{L} = \overline{\psi}(i\gamma^\mu D_\mu - m)\psi = \overline{\psi}(i\gamma^\mu \partial_\mu - m)\psi + q\overline{\psi}\gamma^\mu A_\mu \psi \tag{32}$$

Note that we have a severe implication here: The requirement that the quantum phase can be chosen independtly at all points in spacetime *necessitates* the existence of the gauge field A_μ, *and* it dictates the form of the interaction between the Fermion field and the gauge field (the minimal coupling term) $q\overline{\psi}\gamma^\mu A_\mu \psi$. Avoiding rigor, we can make the identification of this gauge field A_μ with the electromagnetic 4-potential.

Again, note that this transformation does not map between two distinct *physical* states - it simply represents a redundancy in the mathematical representation of the state. All that changes are the non-physical quantities.

3.4.3. Anderson-Higgs Mechanism

In section 3.3.1, I mentioned that when the superfluid state occurs in a liquid made up of electrons, it is called a superconductor. Such an identification can be made, but we must consider the fact that the superconducting state also possesses charge. The frictionless flow of the fluid in a neutral superfluid thus becomes a current that flows without resistance in a superconductor.

Let us begin the analysis by taking the effective Lagrangian of the superfluid Eq. 19, with the interpretation of the complex scalar field remaining the same as that described before in the Ginzburg-landau theory of neutral superfluids. We introduce fluctuations as before, but we ignore the massive excitation field Eq. 22 due to our focus being low-energy excitations. Taking Eq. 23 and remembering that this is the Lagrangian for the *phase*, we require that it be invariant under the local transformation:

$$\theta(\boldsymbol{r}, t) \to \theta(\boldsymbol{r}, t) + \Lambda(\boldsymbol{r}, t)$$

Putting in this local symmetry requires that we deal with the excess terms from the derivatives:

$$\boldsymbol{\nabla}\theta \to \boldsymbol{\nabla}\theta + \boldsymbol{\nabla}\Lambda$$

$$\partial_t \theta \to \partial_t \theta + \partial_t \Lambda$$

To restore invariance, we follow the minimal coupling prescription devised in the previous section, and introduce a gauge field A_μ with the components $(\phi, -\boldsymbol{A})$ - the electromagnetic 4-potential. We take care to replace q, the conserved Noether charge springing from the $U(1)$ symmetry, with e^* - the charge of the particles that couple to the electromagnetic gauge field. Microscopic theory (BCS) reveals that these particles are Cooper pairs, and $e^* = 2e$, where e is the charge of the electron. We will not be representing the expressions in Natural units here, to both respect the fact that the speed of propagation is $c^* \neq c$ and to align with [1]'s treatment of the topic. The altered covariant derivatives are:

$$\boldsymbol{\nabla}\theta \to \boldsymbol{\nabla}\theta - \frac{e^*}{\hbar}\boldsymbol{A}$$

$$\frac{1}{c^*}\partial_t\theta \to \frac{1}{c^*}\partial_t\theta + \frac{e^*}{\hbar}\phi$$

Putting these into the Eq. 23, we obtain the gauged low-energy lagrangian:

$$\mathcal{L}_\psi = \frac{1}{2}\Psi_0^2\left[\left(\frac{\dot{\theta}}{c^*} + \frac{e^*}{\hbar}\phi\right)^2 - \left(\boldsymbol{\nabla}\theta - \frac{e^*}{\hbar}\boldsymbol{A}\right)^2\right] \tag{33}$$

This is not the entire lagrangian, though. Since we have introduced the gauge field to agree with special relativity, the gauge field itself must act as a dynamical entity i.e. it must have its own equations of motion. In this case, that would be the Maxwell equations. This is typically included via an electromagnetic lagrangian $\mathcal{L}_{EM}$ that contains a kinetic term with a square of the field strength tensor as defined in Eq. 27. Represented in terms of physically observable fields $\boldsymbol{E}$ and $\boldsymbol{B}$, this term is:

$$\mathcal{L}_{EM} = \frac{1}{2\mu_0}\left[\left(\frac{\boldsymbol{E}}{c}\right)^2 - \boldsymbol{B}^2\right] \tag{34}$$

Our complete Lagrangian is thus: $\mathcal{L} = \mathcal{L}_\psi + \mathcal{L}_{EM}$

To proceed from here, we could expand the squared terms in Eq. 33, but a more elegant method would be to *redefine* our potentials:

$$\boldsymbol{A} \to \boldsymbol{A} + \frac{\hbar}{e^*}\boldsymbol{\nabla}\theta \tag{35}$$

$$\phi \to \phi - \frac{\hbar}{e^* c^*}\partial_t\theta \tag{36}$$

to essentially "absorb" the derivatives of the phase θ in Eq. 33. Before we proceed note that the Eq. 35 transformation mixes the transverse and longitudinal components. We can say this due to the choice of our gauge and the Helmholtz decomposition theorem: Any sufficiently smooth vector field can be represented as a sum of a perfectly rotational and a perfectly irrotational field:

$$\boldsymbol{A} = \boldsymbol{A}_L + \boldsymbol{A}_T$$

With $\boldsymbol{\nabla} \cdot \boldsymbol{A}_T = 0$ and $\boldsymbol{\nabla} \times \boldsymbol{A}_L = 0$. We identify these component field as transverse and longitudinal components respectively. Since our definition of $\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}$ is set - the longitudinal mode is not associated with any physical degrees of freedom - it is "pure gauge". Consequently, a propagating photon (the excitation of this field) has only transverse modes. The $\boldsymbol{\nabla}\theta$ term is known to be associated with a longitudinal (Nambu-Goldstone) mode as discussed before, and thus longitudinal and transverse modes are being mixed here.

Since this is a proper gauge transformation of the potentials, Eq. 34 remains unchanged, and we find:

$$\mathcal{L} = \frac{1}{2\mu_0\lambda_L^2}\left[\left(\frac{\phi}{c^*}\right)^2 - \boldsymbol{A}^2\right] + \frac{1}{2\mu_0}\left[\left(\frac{\boldsymbol{E}}{c}\right)^2 - \boldsymbol{B}^2\right] \tag{37}$$

Where,

$$\lambda_L = \sqrt{\frac{\hbar}{\Psi_0^2 \mu_0 (e^*)^2}} \tag{38}$$

Which is known as the *London penetration depth*. Thus, we have a purely electromagnetic Lagrangian Eq. 37, *with an additional squared term* proportional to $\frac{1}{\lambda_L^2}$ compared to Eq. 34. In "integrating out" the matter terms via the transformations Eq. 35 and Eq. 36, we have created mass in our electromagnetic field.

Apart from simply identifying a squared term in the Lagrangian, it is possible to directly find the dispersion relations. To do this, we have to apply the Principle of Least Action to Eq. 37 to derive the effective Euler-Lagrange equations. It can be shown that the equations take the form of modified Maxwell equations:

$$\varepsilon_0 \boldsymbol{\nabla} \cdot \boldsymbol{E} - \rho = 0 \tag{39}$$

Which is a modified Gauss's Law, with $\rho = -\frac{1}{\mu_0 (c^*)^2 \lambda_L^2} \phi$ and $\varepsilon_0 = \frac{1}{\mu_0 c^2}$.

$$\frac{1}{\mu_0} \left[\frac{1}{c^2} \dot{\boldsymbol{E}} - \boldsymbol{\nabla} \times B \right] + \boldsymbol{j} = 0 \tag{40}$$

Which is a modified Ampere-Maxwell Law, with $\boldsymbol{j} = -\frac{1}{\mu_0 \lambda_L^2} \boldsymbol{A}$. These may be combined via standard vector identities to obtain a differential equation for $\boldsymbol{A}$:

$$\left[\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{1}{\lambda_L^2} \right] \boldsymbol{A} = \left[1 - \left(\frac{c^*}{c} \right)^2 \right] \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{A}) \tag{41}$$

Putting a wave ansatz $\boldsymbol{A} = A_0 e^{\left(\frac{i(\boldsymbol{p} \cdot \boldsymbol{x} - E_{\boldsymbol{p}})}{\hbar} t \right)} \hat{\boldsymbol{e}}$ into Eq. 41, we find two dispersion relations:

- Longitudinal ($\hat{\boldsymbol{e}} \parallel \boldsymbol{p}$):

$$E_{\boldsymbol{p}} = \sqrt{\left(m_A^2 c^2\right)^2 + \left(\boldsymbol{p} c^*\right)^2} \tag{42}$$

- Transverse ($\hat{\boldsymbol{e}} \perp \boldsymbol{p}$):

$$E_{\boldsymbol{p}} = \sqrt{\left(m_A^2 c^2\right)^2 + (\boldsymbol{p} c)^2} \tag{43}$$

With $m_A = \frac{\hbar}{\lambda_L c}$ (mirroring the form of the de Broglie matter wavelength relation).

This is interesting, as we have two dispersion relations for particles *with mass* m_A. The excitations clearly have transverse and longitudinal components, with the former propagating at the speed of light c and the latter propagating with speed c^*.

If there was no coupling between matter and light (as it is in the case of the neutral superfluid), we have massless photons with two transverse components (polarization degrees of freedom) propagating at the speed of light c, and one longitudinal Nambu-Goldstone mode with massless bosons propagating at the speed c^*.

But the coupling between matter and the electromagnetic field has combined these modes, giving us only one mode - usually called a **plasmon**. This mode has massive excitations, with fast transverse components and a slow longitudinal component. This is called the **Anderson-Higgs effect**. The gauge field is said to "swallow" the Nambu-Goldstone mode, leaving behind a single massive bosonic mode.

3.4.4. Consequences of the Anderson-Higgs effect for superconductors

In Quantum Field Theories, forces are understood to be mediated by the exchange of virtual gauge bosons, with the effective range of the force being determined by the mass of this virtual boson. This can be understood using the energy-time Heisenberg Uncertainty Relation:

$$\Delta E \Delta t \sim \hbar \tag{44}$$

This argument lacks rigor, but one can think of a quantum fluctuation as a "temporary particle" that "borrows" some energy ΔE from the vacuum. As per Eq. 44, it can borrow it for a time-scale approximated by:

$$\Delta t \sim \frac{\hbar}{\Delta E}$$

Therefore, a long wavelength/low-energy force carrier boson, generated like this, can only travel a distance:

$$\Delta x \sim c \Delta t \sim \frac{\hbar}{m_A c} = \lambda_L$$

In vacuum, the mass of the photon is $m_A = 0$, and thus this distance diverges: $\Delta x \to \infty$. But inside the superconducting material, $m_A \neq 0$, and thus λ_L is finite. As a result, the magnetic field produced by these photons (inside the superconductor) can only be transmitted a finite distance on the order of λ_L - which is typically on the range of tens to hundreds of nanometers. This means that inside the bulk of the superconductor, the magnetic field vanishes - *Superconductors are perfect diamagnets*. This is called the **Meissner effect**, and it is the defining property of the superconducting state.

3.5. Topological defects in the order parameter

Let us take the example of the Ginzburg-Landau order parameter $\Psi(r)$ as an example. This is a complex scalar field, and we may represent this as:

$$\Psi(\boldsymbol{r}) = |\Psi(\boldsymbol{r})| e^{i\theta(\boldsymbol{r})}$$

Since the phase θ and the magnitude $|\Psi|$ are supposed to be continuous functions of $\boldsymbol{r}$ as a consequence of it being a macroscopic wavefunction, we require that it be *single-valued* at every point.

Thus, if we are to traverse and closed contour $\mathcal{C}$ in space, we must have the value of the magnitude and the phase return to the same value. If we parametrize this closed path in terms of arc-length and normalize - so that the starting point is $d = 0$ and the ending (which is the same as the starting point) point is $d = 1$, we impose the requirement:

$$\theta(d = 0) = \theta(d = 1) + 2n\pi, n \in \mathbb{Z}$$

n is called the *winding number*, and it essentially counts how many times the phase θ "winds around" itself. Since $n \in \mathbb{Z}$, smooth local variations in the phase can never transition the winding number between allowed values - it is effectively fixed by the boundary conditions. Configurations with $n = 1$ are called *votices*, and those with $n = -1$ are called *anti-vortices*.

These defects are termed *topological* because Topology is the mathematical field that concerns itself with the study of properties of geometrical objects that can not be changed by smooth deformations of the objects - such as stretching or bending, but not tearing or gluing.

In the case of superfluid, the naming of "vortex" makes intuitive sense, as at the center of the so-called vortex the magnitude of the order parameter *must go to zero* - i.e. there is no superfluid at the center of a vortex.

To demonstrate this, let us consider the low-temperature symmetry broken phase of the superfluid. Here, the magnitude is non-zero in the bulk due to SSB, and it has the value $\Psi_0 = \sqrt{-\frac{\alpha}{2\beta}}$ as per standard Landau theory. Now, let us assume azimuthal symmetry, and assign the phase as:

$$\theta(\phi) = n\varphi$$

Where n is the winding number, and ϕ is the azimuthal coordinate. Putting this into Eq. 17, we can derive an expression for the velocity:

$$\boldsymbol{v}_s = \frac{\hbar n}{mr}\hat{\boldsymbol{\phi}}$$

Where r is the radial distance from origin. Assuming that the order parameter is fixed at the bulk value Ψ_0, we can now express the kinetic energy density in the superfluid as:

$$\varepsilon_k = \frac{1}{2}m\rho_s\boldsymbol{v}_s^2 = \frac{1}{2}m|\Psi_0|^2\left(\frac{\hbar n}{mr}\right)^2 \propto \frac{n^2}{r^2}$$

Integrating this over a 2D plane (for example) to get the total kinetic energy:

$$E_k \propto \int_0^R \frac{n^2}{r^2} r\,\mathrm{d}r = n^2\int_0^R \frac{1}{r}\,\mathrm{d}r$$

Thus, $E_k \to \infty$ as $r \to 0$, diverging logarithmically due to the singularity at the origin. The only way to prevent this from occuring for some$n \neq 0$ is for the magnitude of the order parameter to deviate from the bulk value Ψ_0 and go to zero at the origin.

Such vortices are found in several materials - ferromagnets, superfluids, superconductors, etc. In 2 dimensions, vortices dominate the low-energy physics of the system and lead to a new kind of phase transition called the *Kosterlitz-Thouless* phase transition which leads to *quasi-long range order*.

These defects have all the properties that can be associated with particles - they can propagate, they have long-range interactions, and they are massive because of the energetic costs associated with supressing the order parameter at the center/core of the defect. Provided that the order parameter has some continuous symmetry, we can expect topological defects to show up.

4. Beyond the standard model of materials

Since the components of the standard model of materials is based on the concepts of *adiabatic continuity* and *order parameters*, we cannot describe states that do not fit into this paradigm. So, if we are to extend our understanding of the myriad phases of matter, we need to start here. Some places we can look for such situations are:

1. Long-range order implies the existence of a non-zero local order parameter.Thus, systems with no long-range order are not well described by the standard model.
2. Frustrated systems, i.e. systems where it is not possible to satisfy all the ground state constraints simultaneously. These systems host exotic phases of matter with no counterpart in the standard model.
3. States of matter where the inter-particle interactions are so strong that it can not be shown to be adiabatically connected to any simpler, easily solvable system.

In the following sections, states of matter where the paradigm of the standard model fails to provide a full description are discussed. Several motifs that show up in these states are:

- Fractionalization of properties: Sometimes the elementary excitations of the phase carry *fractions* of the properties of the usual elementary particles. This may mean electric charge, spin, or other properties.
- Frustrated systems lead to massive degeneracy of the ground state - often leading to the natural description of the ground state being an *emergent gauge field*.
- Many states of *quantum matter* are topologically non-trivial.
- Bulk-boundary correspondence: The topological states in the bulk of the material has consequences on several properties measured at the boundary between the material and vacuum.

4.1. Correlations and Entanglement

Fundamentally, *correlation* is measure of how “aware” one part of a system is of another part of the system. In more concrete terms, it measures how much a change in a degree of freedom in some point in the system influences/is influenced by the changes in a degree of freedom of the system at some other point.

Since statistical information of this nature is encoded in the join probability distribution of the measurable that we are concerned with, evaluated at two different points, the simplest measure of correlation would be the 2nd moment of the joint probability distribution.

Let us take the example of spins interacting with each other on a lattice. We ccan define a spin-spin correlation function:

$$C(r_{ij}) = \langle \boldsymbol{S}_i \cdot \boldsymbol{S}_j \rangle$$

Where i and j denote the lattice points, and r_{ij} is the distance between them. The LHS is the expectation value of the product of the spin values at the two points.

In the low-temperature ferromagnetic phase, we would expect all spins to be aligned - long-range order. In such a case, we have $\lim_{r\to\infty} C(r) = m^2$, where m is the average magnetisation of the system. At high temperatures, the two spins are uncorrelated and we have $\lim_{r\to\infty} C(r) = 0$.

Correlations can be broadly classified into two categories: Classical and Quantum.

4.1.1. Classical Correlations

Classically, two random variables are said to be correlated if their joint probability distribution *does not* factorize into a product of their respective marginal distributions:

$$P(A, B) \neq P(A)P(B)$$

On top of this definition, two extra conditions are imposed on correlated quantitites by classical physics:

- Realism: Physical properties have *definite* values that exist regardless of observation. If we were to measure the spin of an electron, and we find it to be an up spin, classically we would say that the electron was in the up spin state before we measured it.
- Locality: An observation performed on one of the two quantities that are (classically) correlated can not *instantaneously* affect the other quantity's state or measurement outcome.

If we enforce these two conditions in tandem, it would imply that the only possible *cause* for a classical correlation is a shared history of the two correlated quantities. This gives rise to the idea of a *Local Hidden Variable* (LHV) theory, that postulates the existence of a random variable λ such that,

$$P(a, b) = \int \mathrm{d}\lambda \rho(\lambda) P_A(a \mid \lambda) P_B(b \mid \lambda) \tag{45}$$

Where $\rho(\lambda)$ is a classical probability distribution. Thus, the variable λ encodes in its own distribution all the shared information between the two quantities that are classically correlated - it represents the shared history. The consequence of this is that when a value of λ is specified, the joint probability distribution of the classically correlated quantities factorize into marginal distributions:

$$P(a, b \mid \lambda) = P_A(a \mid \lambda) P_B(b \mid \lambda)$$

The systems being measured have thus already "agreed" on how to respond to a measurement, before the measurement occured, and already resides in the required states.

4.1.2. Quantum Correlations

To discuss quantum correlations, it is necessary to switch to a description in terms of *density matrices*. This is because density matrices are able to provide a description that accounts for both statistical uncertainties (i.e. ignorance of the exact state) and quantum uncertainties (i.e. superposition).

Let us consider a bipartite quantum system with subsystems A and B. The states of this system reside in the combined Hilbert space $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$. The most general state of this system can be described as a density matrix ρ_{AB}.

To find a quantum analog of classical correlation described in the previous section, we must construct the density matrix ρ_{AB} of the system using a LHV theory. Appropriate for a quantum system, we construct:

$$\rho_{sep} = \sum_i p_i \rho_A^{(i)} \otimes \rho_B^{(i)} \tag{46}$$

Which is called a *separable state*. This is as if an external source prepares the subsystem A in the local state $\rho_A^{(i)}$ and subsystem B in the local state $\rho_B^{(i)}$ with a classical probability p_i. This is exactly the form taken in Eq. 45, with the integral replaced with a sum.

Quantum correlation is defined by the negation of this description, i.e. states of the system that can not represented like Eq. 46 are said to be quantum correlated.

A popular example of such a state is the spin-singlet state of a system that comprises of two spin-1/2 particles:

$$|0,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle_A \otimes |\downarrow\rangle_B - |\downarrow\rangle_A \otimes |\uparrow\rangle_B)$$

The density matrix of this pure state (i.e. no statistical uncertainties) is $\rho_{AB} = |0,0\rangle\langle 0,0|$, which is:

$$\rho_{AB} = \frac{1}{2}(|\uparrow,\downarrow\rangle\langle\uparrow,\downarrow| - |\uparrow,\downarrow\rangle\langle\downarrow,\uparrow| - |\downarrow,\uparrow\rangle\langle\uparrow,\downarrow| + |\downarrow,\uparrow\rangle\langle\downarrow,\uparrow|)$$

It is not possible to write this down like an LHV Eq. 45, since the coefficients of the off-diagonal elements are not classical probabilities. Since this stae can not be decomposed into a classical statistical mixture of local states, the subsystems A and B do not possess well-defined states independent of each other. They exist as single, indivisible state.

John Bell, in 1964, proved that this mathematical non-separability leads to observable non-classical correlations in quantum systems that have been prepared in certain states with respect to measurements. This is quantified in *Bell's inequality*, the violation of which indicated quantum correlations. The inequality, however, invokes the idea of locality - this is a problem for measuring quantum correlations in certain systems as the natural description of these systems require delocalised states. A detailed discussion of Bell's inequality and related ideas is beyond the scope of this report.

Conventionally, we define *Quantum Entanglement* as a measure of the quantum mechanical correlations that allows for a violation of Bell's inequality.

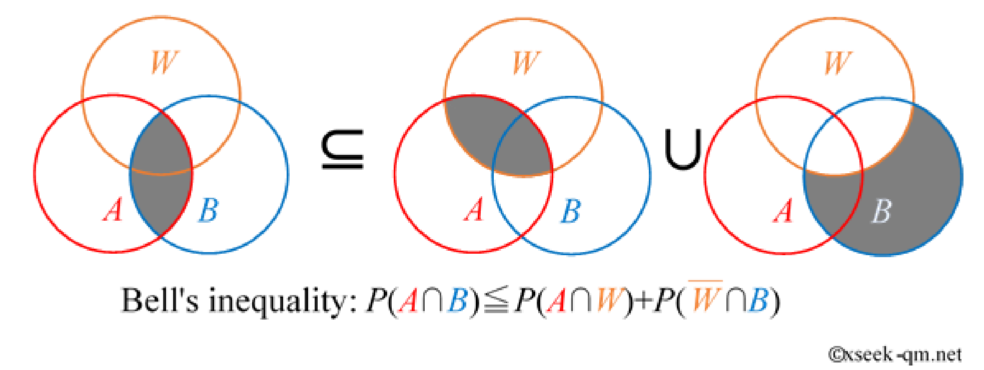

Figure 5: A diagram representing Bell's inequality. Source: seek-qm.net

4.1.3. Hartree-Fock Approximation to measure quantum correlation

The Hartree-Fock state for a system of interacting fermions is defined to be the lowest energy state that can be represented as an antisymmetrized product of single particle wavefunctions - i.e. as a single Slater determinant. This is effectively used as an ansatz state in variational energy minimisation approaches.

Since the HF state is a Slater determinant, it inherently satisfies the Pauli Exclusion Principle, and thus represents correlation between same-spin fermions. But in the process, it entirely ignore the *dynamical correlation* that exists due to interactions between the fermions, particularly in opposite-spin fermions. Due to this, the HF state is a good reference state and the quantum correlations that we are looking to measure must be an improvement over the HF state.

4.1.4. Asymmetric two-site hubbard model

Note that there is a crucial difference between the definitions of Quantum Correlations as presented in Section 4.1.2 and Section 4.1.3:

- If a state is correlated, one can not write down a wavefunction for *any particle* without specifying what the *other particles* are doing.
- If a state is entangled, one cannot write down a wavefunction *anywhere* without specifying what is happening *elsewhere.*

Let us consider a toy model with two electrons in two orbitals to demonstrate this:

$$H = -t \sum_{\sigma} \left(\hat{c}_{1\sigma}^{\dagger} \hat{c}_{2\sigma} + H.c. \right) + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} - \Delta \sum_{\sigma} (\hat{n}_{1\sigma} - \hat{n}_{2\sigma}) \tag{47}$$

Where it is represented in 2nd quantized form, and $\hat{c}_{i\sigma}^{\dagger}$ and $\hat{c}_{i\sigma}$ represents the creation and annihilation of a electron in site $i = 1, 2$ with spin σ. The meaning of the terms is as follows:

- Kinetic/Hopping term: The first term indicates the energy gained by the system due to the delocalization of electrons, dictated by the parameter t.
- Interaction/Correlation term: The second term indicates the energy cost of poutting two electrons in the same site due to on-site coulombic repulsion between them, dictacted by U.
- Asymmetric potential term: The third term applies an asymmetric local potential on each site. Depending on the sign and magnitude of Δ, one site is preferred over the other.

Now, we attempt to find the ground state for three extreme regimes of the parameters (t, U, Δ) that specify the problem:

1. Completely asymmetric limit: Here, $U = t = 0$ and $\Delta = 1$. This means that the electrons do not interact, and do not hop between the sites. The Hamiltonian reduces to:

$$H = -\left(\hat{n}_{1\uparrow} - \hat{n}_{2\uparrow} \right) - \left(\hat{n}_{1\downarrow} - \hat{n}_{2\downarrow} \right) \tag{48}$$

   Due to this, the lowest energy configuration is reached when both electrons are placed in site 1 (where the coefficient is -1). In second quantized form, this state is:

$$|N\rangle = c_{1\uparrow}^{\dagger} c_{1\downarrow}^{\dagger} |0\rangle \tag{49}$$

   Since this is a simple product state (in real space), we can factor this as $|\uparrow, \downarrow\rangle_1 \otimes |0\rangle_2$. This means that there is **no entanglement**. Since the state is generated by a single pair of creation operators acting on vacuum, it is a single Slater determinant and thus the it is **not correlated**.

2. Non-Interacting Symmetric Limit: Here, $U = \Delta = 0$ and $t = 1$. This means that the electrons do not interact and no site is preferred, but they are allowed to hop between the sites. Only the first kinetic term survives:

$$H = -t \sum_{\sigma} \left(\hat{c}_{1\sigma}^{\dagger} \hat{c}_{2\sigma} + \hat{c}_{2\sigma}^{\dagger} \hat{c}_{1\sigma} \right) \tag{50}$$

Since the electrons do not interact, the single-particle Hamiltonian can be diagonalized using (anti)bonding molecular orbitals - $a^\dagger_{\pm\sigma} = \frac{1}{\sqrt{2}}\left(\hat{c}^\dagger_{1\sigma} \pm \hat{c}^\dagger_{2\sigma}\right)$. With the case of a positive t, we have the ground state as:

$$|E\rangle = \frac{1}{\sqrt{2}}\left(\hat{c}^\dagger_{1\uparrow} - \hat{c}^\dagger_{2\uparrow}\right)\frac{1}{\sqrt{2}}\left(\hat{c}^\dagger_{1\downarrow} - \hat{c}^\dagger_{2\downarrow}\right)|0\rangle \tag{51}$$

Which can be combined with fermionic anticommutation relations to get:

$$|E\rangle = \frac{1}{2}\left[\hat{c}^\dagger_{1\uparrow}\hat{c}^\dagger_{1\downarrow} + \hat{c}^\dagger_{2\uparrow}\hat{c}^\dagger_{2\downarrow} - \left(\hat{c}^\dagger_{1\uparrow}\hat{c}^\dagger_{2\downarrow} - \hat{c}^\dagger_{1\downarrow}\hat{c}^\dagger_{2\uparrow}\right)\right]|0\rangle \tag{52}$$

This state can not be factored into a product of states at sites 1 and 2, and thus this is **entangled**. Despite the entaglement, this state was created by putting non-interacting electrons into single particle anti(bonding) states - this this is **not correlated**.

3. The strongly Interacting Limit: Here, $\frac{U}{t} \to \infty$ and $\Delta = 0$. This means that no site is preferred, and the on-site repulsion is much higher than the kinetic energies. Thus, double occupancy states are strongly suppressed. In $S_z = 0$ subspace, this leaves the valid basis states $|\uparrow, \downarrow\rangle$ and $|\downarrow, \uparrow\rangle$. The residual hopping allows the system to lower its kinetic energy by forming a superposition of these states, forming the ground state:

$$|B\rangle = \frac{1}{\sqrt{2}}\left(\hat{c}^\dagger_{1\uparrow}\hat{c}^\dagger_{2\downarrow} - \hat{c}^\dagger_{1\downarrow}\hat{c}^\dagger_{2\uparrow}\right)|0\rangle \tag{53}$$

This can not be factored into local states and is thus **entangled**. Also, this can not be reduced into a single pair of creation operators acting on vacuum either, meaning that this state is **correlated**.

Note that a ground state that is *not entangled, but correlated* is not possible for this model. Whenever the ground state is separable, it can be written as a single Slater determinant.

4.2. Confined/Deconfined; Bound/Free

We begin by defining the terms:

- Confined: Particles that interact with a confining interaction are free at short distances (high energies), but experience a force that grows with distance. It takes an infinite amount of energy to pull pair of confined particles infinitely apart - it is like an elastic band being stretched.
- Bound: Particles that interact via forces that decrease in magnitude with an increase in distance can have *bound* states, where it takes a finite amount of energy to separate them to infinity.

Quarks are an example of confintement - they are never observed in isolation and always come in confined “colorless” combinations. These can be three quarks in a proton, or a quark and an antiquark in a pion, etc.

The electron in its ground state in a Hydrogen atom is in a bound state - it takes approximately $13.6eV$ of energy to ionize the atom.

What is of interest in the study of condensed phases of matter is the fact that we can have situations where particles that are otherwise confined, can become “deconfined” in the material. A popular example of this is discussed in the next section.

4.3. Magnetic Monopoles in Spin ices

4.3.1. Ice rules

There are several different crystalline phases of H_2O - atleast 17 are known. Almost all the ice in the biosphere is in a phase called I_h, which we will focus on here.

In the I_h phase, the H_2O molecules follow so-called "ice rules" to form the crystal - each oxygen atom forms:

- Two short covalent bonds with two Hydrogen atoms
- Two long hydrogen bonds with two Hydrogen atoms

So long as the ice rules are followed, energy is minimized. Linus Pauling showed that there is a *macroscopically large number* of ways to obey the ice rules, and that the ground state of ice is *highly degenerate* as a result of this. The Hydrogen atom positions are highly disordered in this phase. This shows up as a "residual entropy" in ice, which remains even at absolute zero (seemingly in violation of the Third Law of Thermodynamics). Pauling estimated this residual entropy to be:

$$S = \frac{k_B N_H}{2} \ln\left(\frac{3}{2}\right) \tag{54}$$

Where N_H is the number of hydrogen atoms. The vioation of the third law is only apparent, as the I_h phase is not the equilibrium phase at absolute zero.

4.3.2. Spin Ices

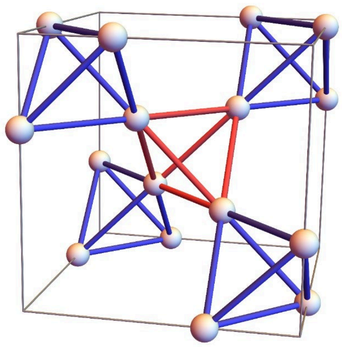

Figure 6: Diagram of pyrochlore lattice. Source: OIST

Spin ices are magnetic materials that consist of magnetic moments that exist on a *pyrochlore lattice*, which can be imagined to consist of the midpoints of the oxygen-hydrogen bonds in I_h ice. We imagine that the centers of the tetrahedra thus formed are occupied with non-magnetic atoms that create a strong crystal field (electrostatic field), that allow for the magnetic moments to point in only two directions - into or out of the tetrahedron.

The interactions between these moments can be minimized when each tetrahedron has two spin pointing in, and two spins pointing out - equivalent to the two "ice rule" configurations of the hydrogen atoms. Thus, Pauling's argument as presented in Section 4.3.1 carries over - this system also has a residual entropy as given in Eq. 54 with $N_H \to N_s$, the number of moments/spins.

An easy way to visualize this system to form a *checkerboard lattice*, which is constructed by taking a top-down projection of the otherwise 3D pyrochlore lattice onto a plane. Restricting ourselves to "one layer", the spins now effectively reside on a square checkerboard-like lattice, with each tetrahedron being represented by the intersection point of the checkers. Certain facts about the physics of the system does change in this, but the facts relevant to this discussion do not.

Let us deal with an Ising model with antiferromagnetic interactions on this lattice, given by the Hamiltonian:

$$H = j \sum_{<ij>} \sigma_i \sigma_j \tag{55}$$

Where $\sigma_i \in \{-1, +1\}$ and $< ij >$ represents a sum over the nearest neighbours in the same tetrahedron.

Let us consider a single tetrahedron. There are $\binom{4}{2} = 6$ interactions, and $2^4 = 16$ possible configurations of the spins on the corners of the tetrahedron. It is *impossible* to minimize all 6 interactions simultaneously - the best that can be done is a minimization of any 4 of the interactions. This means that the system is *geometrically frustrated*. The minimization is achieved for any configuration that adheres to the ice rules, i.e. has a 2-in-2-out configuration. Since this configuration minimizes the energy of each tetrahedron, it will also minimize the energy of the entire lattice.

Let N_s be half the number of tetrahedra (as each spin is shared by two tetrahedra). Then the total number of possible ice state configurations is given by:

$$N_{ice} = 2^{N_s} \left(\frac{6}{16}\right)^{\frac{N_s}{2}} \tag{56}$$

This leads to the Pauling residual entropy,

$$S = k_b \ln(N_{ice}) = \frac{k_B N_s}{2} \ln\left(\frac{3}{2}\right) \tag{57}$$

Like the topic of the article suggests, our interests lie in studying the low-energy excitations of this phase. The simplest way to create these is to introduce a "defect" by flipping a spin, i.e. taking it from a 2-in-2-out configuration to a 3-in-1-out or 1-in-3-out configuration. In our nearest neighbour model, this would cost an energy of $\sim 4J$ due to the 4 interactions this flipped spin has - and it should be viewed as a defect in the two tetrahedra that share this spin. The energy cost indicates outright that there is a gap in this excitation mode - these excitations are massive and require this amount of energy to be formed.

Since the defect has broken the ice rules in the two tetrahedra that it is a part of, we can restore the 2-in-2-out rule on one of them by flipping one of its other spins. This, however, creates another defect in a neighbouring tetrahedron that shares that spin. In effect, this amounts to a defect "moving" or "hopping" to another tetrahedron in the lattice. These moves are all low-energy (at most $\sim 2J$ for a move), and the defects can move freely on this lattice as independent entities. The creation of more defects requires more energy than this hopping, as they are massive. More sophisticated calculations can be carried out to confirm these conclusions, but those are beyond the scope of this report.

The properties of these defects make them relevant to the topic of "deconfinement". In the ground states that follow the ice rule, the net magnetic moment of the system vanishes. Flipping a spin creates a magnetic dipole moment - one pole is at each of the defects created. As described before, these defects can hop around independent of each other on the lattice. These means that the magnetic monopoles here are *not confined to each other*, as we expect in vacuum. **The elementary excitations of a spin ice are magnetic monopoles**. This is also an example of *fractionalization*, as the dipoles (or higher multipoles) are fractionalized into monopoles.

This can be easily extended, and any such geometrically frustrated system with a ground state following ice rules and having a binary degree of freedom demonstrate fractionalization and deconfinement.

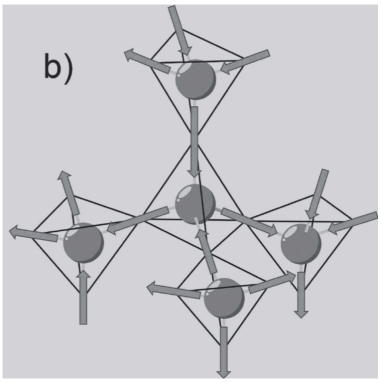

Figure 7: Illustration of the 2-in-2-out Pauling ice rule for a spin ice. Source: [1]

4.4. The Integer Quantum Hall Effect (IQHE)

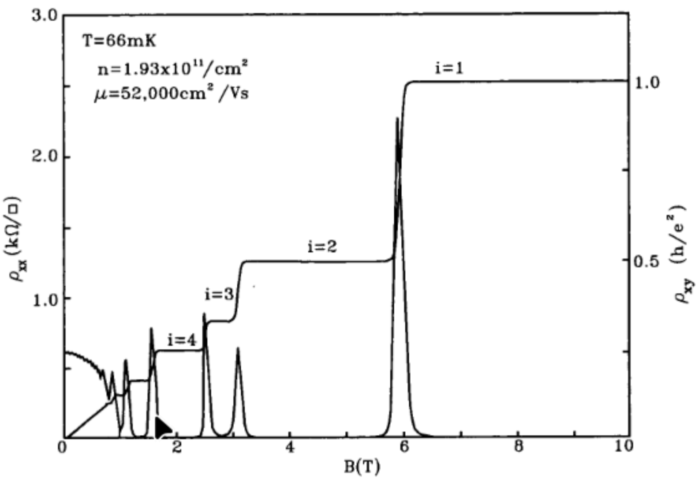

Figure 2. ρ_{xx} and ρ_{xy} of a relatively low mobility 2DEG in GaAs/Al$_x$Ga$_{1-x}$As. The plateaus in ρ_{xy} are quantized in the natural conductance unit e^2/h with integer quantum numbers $i = 1, 2, \ldots$ Data taken by H. P. Wei.

Figure 8: Integer Quantum Hall Effect. Source: [6]

The classical hall effect is seen in conductors where a current flows in, say, the x-direction and a magnetic field of some magnitude B is provided in the z-direction. As a result of this, a potential difference is created in the y-direction - which can be measured at the edges of the conductor. In this case, the effect is easy to understand and requires us to model only the Lorentz force on the electrons, which are imagined to drift at some fixed velocity and scatter every so often. Drude theory can provide us with qualitative results regarding the classical hall effect, and it is observed that the hall potential changes smoothly with the magnitude of B.

Now, in 2-dimensional electron gaes (2DEGs, henceforth), with a current set up in a particular direction in the plane and a homogenous magnetic field perpendicular to the plane, a curious phenomena is observed. Among other things, the plot of the transverse resistivity ρ_{xy} versus the magnetic field magnitude B shows very pronounced *plateaus* at $\rho_{xy} = \frac{h}{e^2}\frac{1}{\nu}$ for integer filling factors ν. These plateaus are so precise, that the so-called von Klitzing constant $\frac{h}{e^2}$ can be measured more accurately than either of the fundamental constants that it consists of.

4.4.1. 2DEGs in perpendicular magnetic field

An electron of charge $-e$ and effective mass m, confined to the $x - y$ plane, is subjected to a uniform static magnetic field $\boldsymbol{B} = B\hat{z}$. The minimal coupling Hamiltonian for the electron is:

$$H = \frac{1}{2m}(\boldsymbol{p} + e\boldsymbol{A})^2 \tag{58}$$

Where $\boldsymbol{A}$ is the vector potential. We have to choose a particular gauge, and we proceed with the Landau Gauge $\boldsymbol{A} = (-By, 0, 0)$ instead of the symmetric gauge as used in [1]. This is mostly due to this gauge supplying a more “visual” description of the wavefunctions, with the wavefunctions being confined to “guiding centers”. This is entirely like the cyclotronic motion of charged particles in magnetic fields.

Using the Landau Gauge, we ariive at the Hamiltonian:

$$H = \frac{1}{2m}(\hat{p}_x - eB\hat{y})^2 + \frac{\hat{p}_z^2}{2m} \tag{59}$$

Note that the x-component of the momentum of the electron $\hat{p}_x$ does not show up in the Hamiltonian and is hence a conserved quantity (i.e. it commutes with the Hamiltonian). Now, we can propose a wavefunction for the electron, which is guessed to be the product of a plane in the x-direction and an unknown function in y:

$$\psi(x, y) = \frac{1}{\sqrt{L_x}} e^{ikx} \phi(y) \tag{60}$$

Where $k = \frac{2\pi n}{L_x}$ is the quantized wavevector in the x-direction, $n \in Z$ is an integer, and L_x is the dimension of the system along the x-direction (in units of length). Acting with the Hamiltonian Eq. 59 on Eq. 60,

$$H = \frac{\hat{p}_y^2}{2m} + \frac{1}{2} m\omega_c^2 (y - y_k)^2 \tag{61}$$

Where $\omega_c = \frac{eB}{m}$ known as the *cyclotron frequency*, and $y_k = \frac{\hbar k}{eB} = k\ell_b^2$, where $\ell_b = \sqrt{\frac{\hbar}{eB}}$ is the *magnetic confinement length*, the fundamental lengthscale of this system. Note that this is exactly the Hamiltonian of a displaced QSHO, and the displacements y_k are said to be located at “guiding centers” of the cyclotronic motion. The energy spectrum is:

$$E_n = \hbar\omega_c \left(n + \frac{1}{2}\right) \tag{62}$$

The usual QSHO spectrum, with $n \in \mathbb{N} \cup \{0\}$. Also note how for for any value of the wavevector k, we have the same energy E_n as it is independent of it. This means that each such *Landau Level* is massively degenerate.

To find the number of available states per Landau level N_Φ, we need to consider the finite dimensions of the 2DEG sample. Allowing for the guiding centers to be fitted inside a sample length of L_y along the y-direction:

$$0 \leq y_k \leq L_y \Rightarrow 0 \leq \frac{\hbar k}{eB} \leq L_y \tag{63}$$

Using the quantisation of $k = \frac{2\pi n}{L_x}$, the maximum value of n can be:

$$N_\Phi = \frac{L_x L_y eB}{2\pi\hbar} = \frac{AB}{\Phi_0} \tag{64}$$

Where A is the crossectional area of the 2DEG, and $\Phi_0 = \frac{h}{e}$ is the magnetic flux quantum.

The macroscopic, continous, 2D density of states of this system has thus collapsed into a set of discrete, massively degenerate energy levels spaced by $\hbar\omega_c$. This does not violate the exclusion principle as the electrons settle into spatially localized wavefunctions in 2-dimensions, around the guiding center y_k.

Now, at $T = 0$, the N electrons in the 2DEG will fill up all possible states upto the Fermi energy E_F, as dictated by the Pauli exclusion principle. Define the *filling factor* ν as,

$$\nu = \frac{N}{N_\Phi} = \frac{N}{AB/\Phi_0} = \frac{n_{2D}}{B/\Phi_0} \tag{65}$$

Where $n_{2D} = \frac{N}{A}$ is the 2-dimensional electron density. Here, ν tells us how many Landau levels are filled at zero temperature. When $\nu \in \mathbb{N}$, the N electrons perfectly fill up the ν lowest Landau Levels, and the $n = \nu$ Landau Level is empty. As per the definition of the Fermi Energy/Chemical potential, one can understand this as the Fermi surface being perfectly in the gap between the $n = \nu - 1$ and $n = \nu$ Landau levels. Note that makes the system *incompressible* - it can no longer respond linearly to applied pressure. In this case, the "pressure" is simply the particle density n_{2D} os the 2DEG, and we are "compressing" the chemical potential μ of the 2DEG. This is because the Fermi surface (i.e. the chemical potential) can no longer shift smoothly with the applied pressure:

- If n_{2D} decreases slightly from an integer ν, then the chemical potential must fall to the $n = \nu - 1$ Landau Level, as any new electron added to this system must reside at that level.
- If n_{2D} increases slightly from an integer ν, the chemical potential must rise sharply into the $n = \nu$ Landau Level, as any new electron can no longer exist in the lower levels.

4.4.2. Edge States and Chirality

In the previous example, the finiteness of the sample was only put in to calculate the degeneracy of a Landau level. However, we must include the effective confining potential of the finite material's edges in the Hamiltonian. Let this potential be $V(y)$, which is flat in the bulk of the material and rises sharply at its edges (say, some $y = -\frac{W}{2}$ and $y = \frac{W}{2}$).

Adding this into Eq. 59, we no longer have states that are strictly degenerate and independent of k. If the potential $V(y)$ varies slowly on the lengthscales of ℓ_B, we can treat it as a perturbation. In the bulk, this simply shifts the energies of the Landau Levels by a constant value (assuming $V(y)$ is flat in the bulk). However, towards the edges the energy bands bend sharply. Roughly,

$$E_{n(k)} \approx \hbar\omega_c\left(n + \frac{1}{2}\right) + V(y_k) \tag{66}$$

Note that the definition of the group velocity of a quantum wavepacket is given as:

$$v_x = \frac{1}{\hbar}\frac{\partial E_n(k)}{\partial k} \tag{67}$$

Using this along with the definition Eq. 66, we obtain:

$$v_x = \frac{1}{eB}\frac{\partial V}{\partial y} \tag{68}$$

Consider the case of the Fermi level sitting inside the gap between the $n = p$ and $n = p + 1$ Landau Levels.

- In the bulk: The energy band is flat, so the electrons are completely localized into cyclotronic orbits. As a result, the carry net zero current, and the bulk is effectively an insulator.
- At the edges: The bent Landau Levels must eventually cross the Fermi level, at which point the LHS in Eq. 68 is non-zero.

Since the slope of the potential $V(y)$ has opposite signs on opposite ends of the 2DEG, the electrons at the "top" edge move in a direction that is opposite to the direction of movement on the "bottom" edge. These are called **chiral edge states**.

4.4.3. Quantization of Hall Conductance

Suppose we drive a new current through the sample by applying a voltage difference across the transverse edges of the 2DEG (i.e. the Hall voltage V_H). This creates a difference in *chemical potentials* between the right-moving states on the "top" edge, and the left-moving states on the "bottom" edge.

$$\Delta\mu = \mu_{top} - \mu_{bottom} = eV_H \tag{69}$$

The net electrical current I is the sum of the currents carried by all the occupied 1D edge "channels". For a single edge channel, the current is given by the charge $-e$ multiplied by the density of states in k-space (which is $\frac{1}{2\pi}$ in this case), and the velocity $v_x(k)$, integrated over the occupied states between the two chemical potentials:

$$I_{single} = -e\int \frac{\mathrm{d}k}{2\pi} v_x(k) = -e\int_{k(\mu_{bottom})}^{k(\mu_{top})} \frac{\mathrm{d}k}{2\pi}\left(\frac{1}{\hbar}\frac{\partial E}{\partial k}\right) \tag{70}$$

Note that the integral nicely cancels out with the derivative, and we are left with:

$$I_{single} = \frac{e^2}{h} V_H \tag{71}$$

Each individual channel thus contributes exactly one quantum of conductance, $(\frac{e)}{h}$ regardless of the specific shape of the confining potential $V(y)$ or the Fermi Energy/velocity.

In the case where the bulk filling factor ν is an integer, there are *exactly* ν Landau Levels that cross the Fermi energy at the edges. This means that there are exactly ν parallel chiral edge channels -

$$I_{total} = \nu \frac{e^2}{h} V_H \tag{72}$$

Which leads to the exact IQHE transverse resistance/conductance expression.

4.4.4. Importance of disorder in IQHE

Note that the analysis carried over the previous few sections were concerned with a hypothetical 2DEG with a perfectly smooth confining potential $V(y)$. The use of a perfect model like this has led us to the required IQHE quantization, but it does not justify *why there are plateaus* - the previous analysis works for exactly $\nu = \frac{N}{N_\Phi}$ filled Landau Levels. Experimentally, we can see the plateaus when a Landau Level has not been completely filled.

The DOS in the previous case consisted of Dirac Deltas centered at the Landau Level energies. Since the filling factor is defined as Eq. 65, any continuous change in the external magnetic field B would lead to a continuous change in the filling fraction, and the Hall conductance would simply form a straight line, inversely proportional to B - and not the plateaus we expect.

The resolution to this is the fact that real systems always have some amount of disorder in them - the confining potential $V(y)$ is *noisy*. In reality, there is a random electrostatic potential landscape $V_N(x, y)$ across the entire sample. Introducing this dosorder potential leads to the lifting of the degeneracy of the Landau Levels - in other words, the Dirac Delta peaks in the DOS widen in finite bands. The central mechanism responsible for the plateaus is called **Anderson Localization** within these broadened bands.

In the bulk, several Landau states are localized near minimas of the disorder potential and thus do not possess the energy required to traverse the potential landscape. These localized states can be thought of as "trapped" electrons, and they do not contribute to the net current. However, at the edges, the Landau Levels are no longer completely flat - the random potential landscape bends the energy levels such that they can cross the Fermi level. The states that cross the Fermi energy at the edges are those that are not localized, and these form the chiral edge states.

So as we reduce the value of B, each Landau Level can hold less electrons and the Fermi Level moves up into the next Landau Level. The lower energy states in this (widened) level are Anderson Localized, and thus can not contribute to the macroscopic current. Consequently, the Hall conductance remains locked at the same level - until the next integer value of ν is reached.

Thus, *disorder is vital for observing the IQHE.*

4.5. Topological order, SPT states, and bulk-boundary correspondence

An important general fact has been highlighted in the discussion on IQHE: the physics of the edge states (which are crucial for the phenomena) are a direct cirect consequence of the bulk properties of the system. This is an example of a general phenomena called *bulk-boundary correspondence.*

Also note that while disorder was required for Anderson localization and the IQHE plateaus to be explained, the exact form of the disorder was not relevant to the final result. This is a characteristic feature of topological states of matter - the physics is robust to the microscopic details of the system. A topological explaination can also justify why the quantization of the Hall conductivity is so precise - no approximate theory can justify this. Thouless, Kohomoto, Nightingale and den Nijs (TKNN) were the first to propose a topological explanation for the IQHE in 1982.

The vacuum is topologically trivial - there is no particular topological character that you can ascribe to it. In contrast, materials in a topologically non-trivial state (like IQHE) must display interesting properties at the edge between the material and vacuum.

One way in which a material can be topologically non-trivial is by displaying some sort of *topological order*. Topologically ordered states do not break any symmetries of the Hamiltonian, unlike the ordered states of Landau Theory. But despite this, there can be no *unitary transformation* that can map them to a topologically trivial state. This suggests that such states are not adiabatically connected to topologically trivial states, and thus must represent a distinct phase of matter.

The only way to achieve topological order is by having *long-range entanglement* in the system, which implies that such order is found in systems where the constituents interact very strongly with each other. As stated before, since no symmetries are broken in this state, a description utilizing local order parameters would not be possible for topologically ordered phases.

Another way of making a material topologically non-trivial is to put it in a *symmetry protected topological* (SPT) state. SPT states are "protected" by some symmetry - it can not be transformed into a topologically trivial state by unitary transformations that *respect the symmetry*. However, just like topologically ordered states, SPT states can also not be described using standard model paradigms like adiabatic continuity and local order parameters. However, unlike topologically ordered states, SPT states *do not require entaglement*.

4.6. Haldane Chain

The 1D Heisenberg chain, also known as the Haldane chain, is an SPT system that shows both fractionalization and bulk-boundary correspondence.

The Hamiltonian for the system is simply:

$$H = J \sum_i \hat{\boldsymbol{S}}_i \cdot \hat{\boldsymbol{S}}_{i+1} \tag{73}$$

Where $S_i = \left(S_i^x, S_i^y, S_i^z\right)$ represents the spin operators for the i-th site, and J represents the exchange coupling.

Focusing on the Wess-Zumino term (see Eq. 26), Haldane provided a topological argument for his set of *conjectures* concerning a antiferromagnet exchange ($J > 0$) 1D Heisenberg chain:

1. For Half Integer chains ($S = 1/2, 3/2, \ldots$): The ground state is gapless, and the spin correlations decay according to a power-law (i.e. algebriacally) -

$$\langle \boldsymbol{S}_0 \cdot \boldsymbol{S}_r \rangle \sim (-1)^r \frac{1}{r^\eta} \tag{74}$$

   For some η. This means half-integer spin chains have *quasi long-range order*.

2. For Integer chains ($S = 1, 2, \ldots$): The ground state has a finite gap Δ with the first excited state, and the spin correlations decay exponentially -

$$\langle \boldsymbol{S}_0 \cdot \boldsymbol{S}_r \rangle \sim (-1)^r e^{-\frac{r}{\xi}} \tag{75}$$

For some ξ correlation length, and Δ is called the Haldane gap. This means integer spin chains have *short-range order*

4.6.1. The AKLT Construction

To prove rigorously that a uniform integer-spin chain can exhibit a spectral gap and exponential decay of correlations, Affleck, Kennedy, Lieb, and Tasaki constructed an exact, solvable model in 1987. This is known as the Affleck-Kennedy-Lieb-Tasaki (AKLT) model.

Unlike the Heisenberg model discussed before, the AKLT model is not a model of interacting spins on a lattice. Rather, it can be thought of as being constructed in the following way:

1. Start with a finite chain of spin-1 particles, each site indexed by an integer i.
2. Represent each spin-1 using two fictitious spin-1/2 degrees of freedom. Let us denote these variable spin-1/2s as left and right - $|\sigma\rangle_{i,l}$ and $|\sigma\rangle_{i,r}$. The physical Hilbert space of a spin-1 particle consists of the basis states $\{|1,1\rangle, |1,0\rangle, |1,-1\rangle\}$. These can be represented as the symmetric triplet states of 2 spin-1/2s (See Section 3.2.3).
3. Take the right spin of site i and the left spin of site $i+1$ and couple them together to form a to construct a spin-singlet state, called a Valence Bond wavefunction:

$$|VB\rangle = \prod_{i=1}^{N-1} \frac{1}{\sqrt{2}} \left(|\uparrow\rangle_{i,r} |\downarrow\rangle_{i+1,l} - |\downarrow\rangle_{i,r} |\uparrow\rangle_{i+1,l} \right) \tag{76}$$

4. To restrict ourselves to the spin-1 subspace (note that the tensor product space of two spin-1/2s is 4-dimensional, whereas the space for a spin-1 is 3-dimensional) we project $|VB\rangle$ onto a state where there is a triplet state/spin-1 on each site, and constructing the AKLT ground state:

$$|AKLT\rangle = \sum_{i=1}^{N} \left(|1\rangle_i \langle\uparrow|_{i,l} \langle\uparrow|_{i,r} + |0\rangle_i \frac{\langle\uparrow|_{i,l}\langle\downarrow|_{i,r} + \langle\downarrow|_{i,l}\langle\uparrow|_{i,r}}{\sqrt{2}} + |-1\rangle_i \langle\downarrow|_{i,l} \langle\downarrow|_{i,r} \right) |VB\rangle \tag{77}$$

5. Now, we need to construct the AKLT Hamiltonian, which has $|AKLT\rangle$ as the exact ground state. Let us start by considering two adjacent sites, i and $i+1$. The total spin of this pair is $\boldsymbol{S}_{tot} = \boldsymbol{S}_i + \boldsymbol{S}_{i+1}$, and thus $\boldsymbol{S}_{tot} \in \{0, 1, 2\}$. Due to the construction of the Valence Bonds, one spin-1/2 is consumed from each site to form a total spin-0 singlet. This leaves only two free spin-1/2s associated with the two sites, one on the extreme left $|\sigma\rangle_{i,l}$ and one on the extreme right $|\sigma\rangle_{i+1,r}$. Two spin-1/2 can couple to form a maximum of total spin 1, so it is not possible in the Valence Bond state for two adjacent sites to have a total spin of 2. With this information, we can construct a Hamiltonian that imposes an energy cost for any configuration where adjacent sites have a total spin 2:

$$H_{AKLT} = J \sum_i \Pi^{(2)}_{i,i+1} \tag{78}$$

Where $\Pi^{(2)}_{i,i+1}$ is the projection operator onto the spin-2 subspace of the pair of sites i and $i+1$.
6. Since $\Pi^{(2)}_{i,i+1}|AKLT\rangle = 0 \forall i$, by definition, $|AKLT\rangle$ is the exact ground state of the AKLT Hamiltonian, with energy $E = 0$.

Now that we have constructed the model starting from designing a ground state and then constructing the Hamiltonian using it, we can express the Hamiltonian in terms of local spin operators:

$$H_{AKLT} = J \sum_i \left[\boldsymbol{S}_i \cdot \boldsymbol{S}_{i+1} + \frac{1}{3} (\boldsymbol{S}_i \cdot \boldsymbol{S}_{i+1})^2 \right] \tag{79}$$

Which defines a bilinear-biquadratic model, where the Hamiltonian can be written as:

$$H_{bb}(\beta) = J \sum_i \left[\boldsymbol{S}_i \cdot \boldsymbol{S}_{i+1} + \beta (\boldsymbol{S}_i \cdot \boldsymbol{S}_{i+1})^2 \right] \tag{80}$$

Clearly, the AKLT Hamiltonian Eq. 79 is $H_{bb}\left(\frac{1}{3}\right)$. It can be shown in the more general H_{bb} model that there no phase transition possible in the region $\frac{1}{3} \geq \beta \geq 0$. Therefore, the principle of adiabatic continuity assures us that the key qualitative predictions of the AKLT model will hold for spin-1 Heisenberg chains (which is a $H_{bb}(0)$ model), i.e. they are adiabatically connected.

This allows us to use the exactly solvable AKLT model to confirm several properties stemming from Haldane's analysis:

- The AKLT state is gapped, as an excitation requires us to break one of the valence bonds (i.e. adjacent site spin singlet). This can cost $\sim J$ energy as it can cause a total spin 2 in adjacent sites.
- Note that the states for $|\sigma\rangle_{1,l}$ and $|\sigma\rangle_{N,r}$ have not been specified in the construction - $|AKLT\rangle$ is the ground state of this model regardless of what value they take. As a result, the ground state has a degeneracy of 4. **This is an example of bulk-boundary correspondence and fractionalization**, as the emergent spin-1/2s at the boundary are due to the valence bonds in the bulk, consisting entirely of spin-1s.
- If we tweak the construction rules for $|AKLT\rangle$we can demonstrate the degeneracy of the ground state is entirely dependent on the topology of the lattice. Suppose we fuse the edge states to form another spin singlet state, i.e. we construct a new ground state:

$$|AKLT'\rangle = \frac{1}{\sqrt{2}} \left(|\uparrow\rangle_{1,l} |\downarrow\rangle_{N,r} - |\downarrow\rangle_{1,l} |\uparrow\rangle_{N,r} \right) |AKLT\rangle \tag{81}$$

  This state is no longer four-fold degenerate. Such a dependence of the ground state degeneracy on the topology of the material is typical of topological matter.
- It can also be shown that the spin-spin correlation length is finite as stated by Haldane, but that is beyond the scope of this report.

Note the important of having quantum entanglement in the construction of the AKLT state - the spin singlet state in the valence bonds is a quintessential entagled state. This sets it apart from IQHE and Spin Ices (other systems with bulk-boundary correspondence and fractionalization), where entanglement is not required. However, the entanglement is short range (between adjacent sites), and thus *this is an SPT state and not topological order* (which requires long range entanglement).

Since the AKLT model is adiabatically connected to the 1D Heisenberg chain, the latter is also an SPT state. Generally, only short range order is possible in 1D, and topological order can thus show up only in higher dimensions.

4.7. Majorana fermions

The natural extension of the phases observed in spin-chains is looking at systems with no specified *local* degrees of freedom, such as systems with itinerant fermions. The main questions become:

- Is it possible to find quantum phases of matter in such systems?
- Do these materials show bulk-boundary correspondence?
- Is it possible to have fractionalized excitations in these materials?

The answer to all three of these questions have been shown to be “yes”, and to see why we need to discuss *Majorana fermions*: A hypothetical fermion that is its own antiparticle. Such fermions must be massless. It is believed that such fermions can emerge in condensed matter systems, and a way to understand the basic physics of these particles is to look at a toy model.

4.7.1. Kitaev chain

This is a toy model for a spin-less p-wave superconductor in 1D. It consists of a chain of N lattice sites that are described by the Hamiltonian:

$$H = -\mu \sum_{i}^{N} \hat{c}_i^\dagger \hat{c}_i - \frac{1}{2} \sum_{i}^{N-1} \left[t \hat{c}_i^\dagger \hat{c}_{i+1} + \Delta e^{i\theta} \hat{c}_i \hat{c}_{i+1} + H.c. \right] \tag{82}$$

Where $\hat{c}_i$ and $\hat{c}_i^\dagger$ are the fermionic annihilation and creation operators at site i. μ is the chemical potential, and t is the hopping integral between neighbouring sites. $\Delta e^{i\theta} \propto \Psi$, which is the Ginzburg-Landau order parameter for the superconducting/superfluid phase discussed in Section 3.3.1. Note that since the fermions are spinless, the exclusion principle prohibits double occupancy at any site.

We now do a something similar to the first step of the AKLT construction, and we represent each spinless fermion with two Majorana fermions. We do this via the transformation:

$$\hat{c}_i = e^{-i\frac{\theta}{2}} \frac{\hat{\gamma}_{r,i} + i\hat{\gamma}_{l,i}}{2} \tag{83}$$

Provided that the Majorana operators $\hat{\gamma}_{\alpha,\beta}$ follow the canonical fermionic anticommutation relations:

$$\left\{ \hat{\gamma}_{\alpha,i}, \hat{\gamma}_{\beta,j} \right\} = \delta_{i,j} \delta_{\alpha,\beta} \tag{84}$$

Notice that $\hat{\gamma}_{\alpha,i}^\dagger = \hat{\gamma}_{\alpha,i}$ as a consequence of this, i.e. creating a Majorana fermion is the same as annihilating one. This representation is valid, as it preserves the anticommutation relations of the underlying spinless fermion operators. I this Majorana basis, the Hamiltonian of the Kitaev chain becomes,

$$H = -\frac{\mu}{2} \sum_{j}^{N} (1 + i\hat{\gamma}_{r,j}\hat{\gamma}_{l,j}) - \frac{i}{4} \sum_{j}^{N-1} \left[(\Delta - t)\hat{\gamma}_{l,j}\hat{\gamma}_{r,j+1} + (\Delta + t)\hat{\gamma}_{r,j}\hat{\gamma}_{l,j+1} \right] \tag{85}$$

So, the problem is specified by values of (μ, t, Δ).

Let us look at the two phases that this model can exhibit, both of which are adiabatically connected to limiting cases of the parameters that are trivially solved.

- $t = \Delta = 0, \mu < 0$: Turning off hopping and pairing, the Majorana operators pair up on each site to recover the original spinless fermion, i.e. we have terms like $\hat{\gamma}_{r,j}\hat{\gamma}_{l,j}$ for each site j in the Hamiltonian. This phase requires energy for the addition of particles, and thus has a gap of $|\mu|$. It is topologically trivial, and it corresponds to the vacuum state for the original fermions.
- $\mu = 0$, $\Delta = t \neq 0$: The hopping parameter is equal to the pairing parameter, The Hamiltonian simplifies drastically to the form:

$$H = \frac{it}{2} \sum_{j=1}^{N-1} \hat{\gamma}_{r,j}\hat{\gamma}_{l,j+1} \tag{86}$$

  Meaning that there is *exclusive* inter-site pairing of Majoranas, between adjacent sites. This is quite like the valence bonds in the AKLT model.

Let us look at the second phase of the Kitaev chain. Looking at the inter-site Majorana pairings, we change basis again and represent the Hamiltonian in this limiting case using the transformation:

$$\hat{d}_i = \frac{\hat{\gamma}_{r,i+1} + i\hat{\gamma}_{l,i}}{2} \tag{87}$$

Which represents a new set of fermions. In this basis, the Hamiltonian simplifies to:

$$H = -t \sum_{j=1}^{N-1} \left(\hat{d}_j^{\dagger}\hat{d}_j - \frac{1}{2} \right) \tag{88}$$

which has a bulk gap of $|t|$.

What is interesting is that there are two Majoranas at the ends of the chain $|\gamma\rangle_{l,1}$ and $|\gamma\rangle_{r,N}$, that do not show up in the Hamiltonian of this phase - they create a degeneracy of 2 in the ground state. It costs nothing to occupy these states.

The existence of these Majorana fermions at the ends are due to the bulk, demonstrating bulk-doundary correspondence. In addition, this is also clearly a case of fractionalization, as the original spinless fermions are fractionalized into Majorana fermions. Note the parallels with the AKLT construction and analysis preceeding this. The two edge Majoranas can be thought of a highly non-local Dirac fermion

The two degenerate ground states of the Kitaev chain have immense potential for use as *qubits* in quantum computation, as they are highly non-local objects - separated by the entire length of the chain. This makes them very resilient to noise, which makes for a good qubit.

Using Bogoliubov-de Gennes momentum space representation, it is possible to write down what the topological invariant is that classifies the two phases of the Kitaev chain. I will not attempt to reproduce this in this report, as it beyond the scope.

4.8. Anyons

In 2-dimensions, Majoranas gain more interesting properties as compared to 1D. Apart from the familiar bosonic/fermionic statistics, in 2D one can have two more families of solutions to the normalization condition of particle exchange. Let us work with a state $|\boldsymbol{r}_1, \boldsymbol{r}_2\rangle$ with two quantum particles, labeled by their position r_1 and r_2 in 2D. The two new "particle statistics" are as follows:

- *Abelian anyons*, where:

$$|\boldsymbol{r}_2, \boldsymbol{r}_1\rangle = e^{i\theta}|\boldsymbol{r}_1, \boldsymbol{r}_2\rangle \tag{89}$$

  With $\theta \neq n\pi$ where $n \in \mathbb{Z}$.

- *Non-Abelian anyons*, where:

$$|\boldsymbol{r}_2, \boldsymbol{r}_1\rangle \neq |\boldsymbol{r}_1, \boldsymbol{r}_2\rangle \tag{90}$$

  i.e. they are fundamentally different states.

These new statistics on exchange as possible precisely because the particles are restricted to 2D. In 3D, a double exchange, i.e. exchanging a pair of particle twice, is physically the same as not exchanging them at all. This restricts us to bosonic and fermionic statistics. But in 2D, due to the topological nature of the paths that exchanging particles need to take, it becomes possible to have these two new exchange statistics.

A 2D generalization of the Kitaev chain is possible, and it is called the *$p + ip$ superconductor*. In such systems, we find that the Majorana fermions are *bound to vortices* (see Section 3.5). If we have an even number of Majoranas in such a system, they can be paired up (arbitrarily) to create fermions - each such fermion has zero energy. If there are $2N$ Majoranas, this gives rise to a 2^N-fold degeneracy in the ground state.

Returning to the focus of this section, we wish to exchange (adiabatically) two such vortex-bound Majoranas. In general, one such exchange will change the state of state of *one* of the other 2^N ground states of the system. Since these are eigenstates, they can not simply differ by a phase - implying that these Majoranas act as Non-Abelian anyons.

4.8.1. Particles bound to vortices

The idea of composite particles that consist of a particle bound to a vortex is in general quite important.

Let us work with a toy model - two bosons with charge q bound to vortices in a plane. We can assume these two to be point particles, and let us place one at the origin.

Now, the vector potential $\boldsymbol{A}$ produced due to the flux at the origin extends beyond the bound state there. If the boson moves on a path that encloses the vortex at the origin, we accumulate a non-zero Aharanov-Bohm (AB) phase.

Now, let us look at exchanges between the quasiparticles. The vortices themselves satisfy bosonic statistics and nothing changes if they are exchanged. The same is true for the exchange of the bosons. But, if we want to exchange the entirety of the bound states themselves, we have a different situation. Note that this will *require* that we choose a clockwise/anticlockwise rotation, and rotate each boson halfway (i.e. $\frac{\pi}{2}$) around the vortex that it is *not bound to*. This causes each composite particle here to pick up an AB phase, and the total state thus picks up a net phase on exchange. These, thus, act like *Abelian anyons*.

This would not work in 3-dimensions, as the clockwise and anticlockwise rotations are the same and can be continuously mapped to each other via rotations in 3D space. This brings back the requirement for the single-valued nature of the wavefunction, resulting in fermionic/bosonic statistics being the only two options.

4.9. Emergent gauge fields

The Haldane and Kitaev chains discussed before are both 1D, and fracionalization is very common in 1D systems. But, it is also possible to have fracionalization in higher dimensional systems. These often require more phenomenology to justify.

Typically, fractionalized particles arise together with *emergent gauge fields*. This is similar to the case of the electromagnetic field accompanying charged particles.

4.9.1. Emergent gauge fields in spin ices

Classical spin ices are good examples of emergent gauge fields. These fields arise in coarse-grained theories of these materials, owing to to the large degeneracy.

As discussed in Section 4.3.2, let us work with a nearest neighbour (classical) Heisenberg model on a pyrochlore lattice. We define the net spin flux at a tetrahedron as:

$$\boldsymbol{L}_\alpha = \sum_{j \in T_\alpha} \boldsymbol{S}_j \tag{91}$$

Where T_α is a set that contains the lattice points in the α-th tetrahedron in the pyrochlore lattice. The lattice formed by these tetrahedra themselves (indexed by α) forms the *dual lattice* of the pyrochlore, and it is a bipartite diamond lattice.

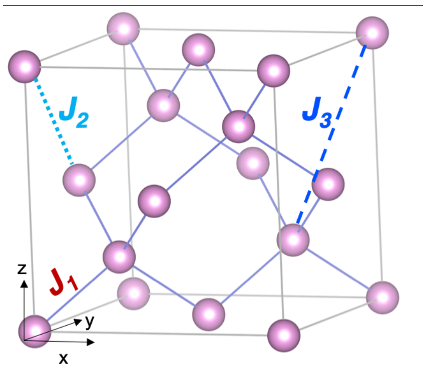

Figure 9: A diamond lattice, the dual of a pyrochlore lattice. Source: ResearchGate

Thus, upto a constant term, we have:

$$H = J \sum_\alpha \sum_{i \neq j, i, j \in T_\alpha} \boldsymbol{S}_i \cdot \boldsymbol{S}_j = \frac{J}{2} \sum_\alpha \boldsymbol{L}_\alpha^2 \tag{92}$$

States satisfying the 2-in-2-out ice rules give us $\boldsymbol{L}_\alpha = 0 \forall \alpha$, and are thereby the ground states of the system. There ae several such states, and the local constraint of $\boldsymbol{L}_\alpha = 0$ can be enforced via a *gauge field*.

To construct such a field, we take a volume element $V(\boldsymbol{r})$ that is large compared to individual tetrahedra and small compared to the sample size, and map the sum of all the spins in this volume element to some coarse-grained field $\boldsymbol{B}(\boldsymbol{r})$:

$$\sum_{i \in V(\boldsymbol{r})} \boldsymbol{S}_i \to \boldsymbol{B}(\boldsymbol{r}) V(\boldsymbol{r}) \tag{93}$$

Consider one of the states where the local constraint is met everywhere. If we were to flip a spin in this state, to maintain the flux constraint we also need to flip another spin of an opposite value on the same tetrahedron. Since each spin is shared by two tetrahedra, this process of flipping spins to satisfy the local flux constraint must be done over a *closed loop* on the lattice.

Generally, flipping spins over a loop that spans the entire system will change the value of $\boldsymbol{B}(\boldsymbol{r})$, but flipping them over small, contractible loops do not affect the coarse grained field.

Let $\mathcal{N}(\boldsymbol{B})$ represent the number of spins in the system that can be flipped without changing the value of the coarse-grained field $\boldsymbol{B}$. For $\boldsymbol{B} = 0$, $\mathcal{N}$ takes its largest value. As $\boldsymbol{B}$ increases, twothings happen:

1. $\mathcal{N}(\boldsymbol{B})$ decreases as the system gets saturated. The system becomes polarized and most of the spins start pointing in the same direction.
2. $\mathcal{N}(\boldsymbol{B})$ decreases as the number of possible closed loops also decrease. This is because, by virtue of being closed, each such loop must have half of its spins point in the other direction of the coarse-grained field.

Thus, there are many spins that may be flipped without changing $\boldsymbol{B}$ when $\boldsymbol{B} = 0$, but very few at saturation. The specific entropy of the system is maximal at zero field.

It is possible for us to obtain analogs to Gauss' Law for magnetism, with this coarse grained field $\boldsymbol{B}$. For the ground states, we have Gauss' Law exactly:

$$\boldsymbol{\nabla} \cdot B = 0 \tag{94}$$

However, with increasing defects (which, as discussed in Section 4.3.2 are magnetic monopoles), the RHS becomes non-zero. It is possible to show that these monopoles follow an analog of Coulomb's Law, but for magnetic charges. As a consequence, this phase is called the *Coulomb Phase*. This field $\boldsymbol{B}$ thus has the gauge freedom of the electromagnetic field, and the entire theory remains invariant under gauge transformations of the underlying vector potential.

4.10. Quantum Spin Liquids

Liquids occupy an intermediate position between solids and gases, as the correlations between the constituents are not strong enough to dominate and establish long range order as in the solids, but neither is it weak enough to be ignore like in the case of the ideal gas. As established in the sections on SSB, there is a deep understanding of long-range order available to modern condensed matter physicists - this is not true for the case of liquids.

States in magnetic materials where there are *strong correlations* between the spins, but no long-range order are called *spin liquids*. A definition that suffices for our purposes is this - a spin liquid is any state that has:

1. Well defined local moments (i.e. spins)
2. Does not have long range magnetic order
3. Can not be adequately described using a model of weakly interacting spins.

Reducing the dimensionality of the magnetic system under study appears to be a straightforward way to supress the formation of long-range order. But the definition of spin liquid is contentious, and often 1D spin chains are excluded from the definition of spin liquids. However, including quantum effects allows for several kinds of spin liquids in more than 1-dimension - a classification of such states is still under progress.

Spin liquids fall outside of the ambit of the standard model of materials because:

- They break no symmetry of the Hamiltonian, and thus can not be described using a local order parameter.
- The correlations of their consituents are strong enough to prevent a spin liquid state from being adiabatically connected to a set of non-interacting spins.

Instead, they are often described using emergent gauge fields and fractionalized quasiparticles.

4.10.1. Dimer Models and RVB states

A geometrically frustrated system, like a triangular lattice with antiferromagnetic exchange or spin ices, naturally supress long-range order due to the ground state being formed via the satisfaction of *local* rules.

The ground state of a two-site antiferromagnet is the spin singlet $|0, 0\rangle$, as we know. The singlet has the additional property of being a *maximally entangled state*: If a spin is in a singlet state, it can not be entangled with any other spin(s). A very useful consequence of this fact is that if we can specify a "pattern" or configuration over a lattice consisting of spin-1/2s, we completely specify the quantum state of the system. Just like the AKLT construction in Section 4.6.1, such states are called *valence bond states*, and the singlets are called *dimers*.

For any given lattice, there could in general be several such patterns or coverings of dimers. Anderson suggested that a spin liquid can be constructed using a quantum superposition of different valence bond states - this is known as a *resonating valence bond state* (RVB).

In addition to potentially being a building block of spin liquids, RVB states also have the potential to provide an alternate description of classical spin ices. Apart from the standard 2-in-2-out ice rules, other rules such as 3-in-1-out or 1-in-3-out can be represented by different coverings of dimers on the dual lattice. These Quantum Dimer models are very useful in constructing low-energy quantum dynamical theories of spin ices, allowing for a convenient description of tunneling between different ice configurations.

4.10.2. Rokhsar-Kivelson Model

Let us consider a square, bipartite lattice with sites A and B, as usual, occupied by spin-1/2 particles at each site. Inter-sublattice interactions are allowed, but intra-subtlattice interactions are not allowed (nearest neighbour interaction scheme).

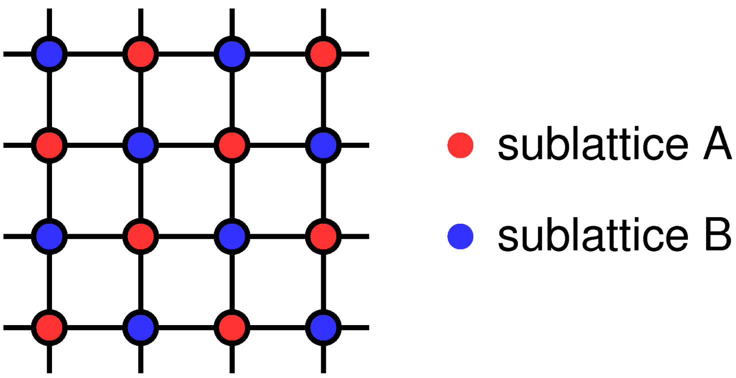

Figure 10: The square lattice with sites A and B. Source: Boston University

A valid dimer covering of this lattice under a 3-in-1-out or 1-in-3-out rule would have to follow certain rules:

- Every site must be connected to *exactly one* dimer. Since a dimer is a spin-singlet, this would completely specify the quantum state of the lattice.
- No site can be left empty.
- No site can share *more than one* dimer. This is a bit redundant, and is essentially a restatement of rule 1.

If we start off with a *finite* lattice, we would want to avoid the effects of the finite edge. The standard way to do away with finite edge effects is to apply a *periodic boundary condition* (PBC). In the case of the square lattice, this would mean that we *wrap around* the x- and y-coordinates, effectively putting the lattice onto a *torus*.

A torus is, in terms of topology, a $g = 1$ surface. Here, g denotes the *genus* of the surface - a fundamental topological invariant. It is essentially a count of how many "holes" a surface has. A $g = 0$ surface, for example, is topologically the same as a sphere, while a $g = 2$ surface is like a scissor, or a cup with two handles.

The genus is important, as it dictates what kinds of curves on the surface can be smoothly deformed and contracted to a point. On a $g = 0$ (spherical) surface, *any* curve can be deformed to a point. But note that in a $g = 1$ (torus), any curve that "winds around" either of the two radii (inner loop, or the outer loop) cannot be contracted to a point. The presence of the hole creates this distinct effect.

Now, on this square lattice we identify each 2×2 square grid as a *plaquette*. Since the diagonally opposite vertices of the square lattice are on the same lattice, and we are restricted to the dimer coverings that follow the rules prescribed previously, an appropriate choice of plaquettes will lead us to two different kinds of them. Due to typographical reasons, I will define my notations as follows:

1. Left-right dimer plaquette: $|\|\rangle_p$
2. Up-down dimer plaquette: $|=\rangle_p$
3. Unflippable plaquette: This has no representation (as it is not required), but physically it represents a plaquette that has atleast one dimer going out of the plaquette.

$|\|\rangle$ and $|=\rangle$ states are called "flippable", as we can go between them without affecting neighbouring plaquettes.

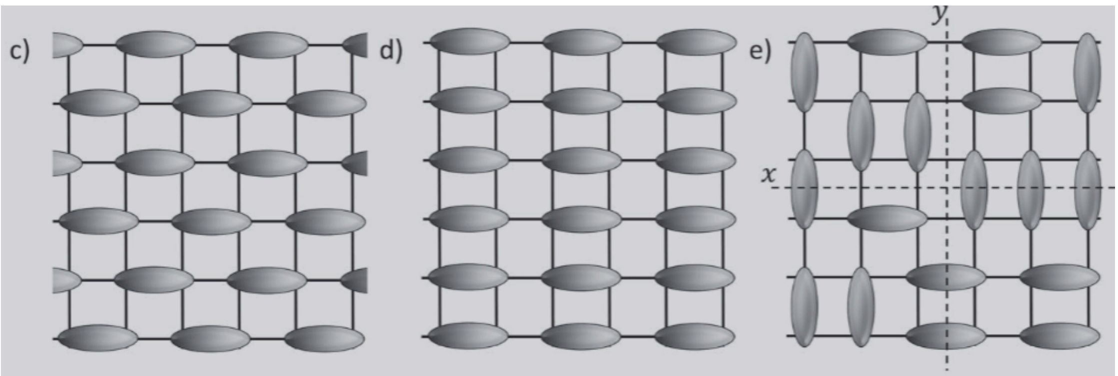

Figure 11: Different dimer coverings, with the grey gradient ovals present on bonds with dimers. (c) depicts the "staggered" phase which minimizes flippable plaquettes. (d) depicts the "columnar" phase, which maximizes flippable plaquettes. (e) demonstrate how reference lines are drawn. Source: [1]

These indicate where the dimers are in the plaquette, and p is the plaquette index. In our representation (due to the covering rules), a local plaquette basis is adequate for describing the state of the system.

Now, we define the Rokhsar-Kivelson (RK, henceforth) Hamiltonian:

$$H = \sum_p \left[-J \left(|\Vert\rangle_p \langle = |_p + | = \rangle_p \langle \Vert |_p \right) + V \left(|\Vert\rangle_p \langle \Vert |_p + | = \rangle_p \langle = |_p \right) \right] \tag{95}$$

There are two distinct terms here:

1. 1st term: This is the *kinetic term*, and it causes flips between the two distinct dimer coverings. The rate of this flip is dictated by the coupling strength J. We will restrict ourselves to systems with $J > 0$, as that promotes flipping, and hence the superpositions that we would expect from an RVB model.

- 2nd term: This is the *potential term*, and effectively counts the number of plaquettes with parallel spins of either kind.

These two terms compete, and the two extreme limits are:

- $\frac{V}{J} \to \infty$: The system seeks to minimize the number of flippable plaquettes, resulting in a staggered phase.
- $\frac{V}{J} \to -\infty$: The system seeks to maximize the number of flippable plaquettes, resulting in a columnar phase.

But something interesting happens when we set $V = J$. The Hamiltonian Eq. 95 can be easily shown to become:

$$H = \sum_p \left[J \left(|\Vert\rangle_p - \langle = |_p \right) \left(|\Vert\rangle_p - \langle = |_p \right) \right] \tag{96}$$

This condition defines a point named the "Rokhsar-Kivelson point", and it is clear that the Hamiltonian has become a *projection operator for an antisymmetric state*. Since it is a sum of projection operators, it is positive semi-definite and thus must have a lower energy bound of $E = 0$ for any possible state.

We intend to find the RK ground state $|\Psi\rangle_{RK}$. For this, the state must be annihilated by *every* local projection operator $P_p = \left(|\Vert\rangle_p - \langle = |_p \right) \left(|\Vert\rangle_p - \langle = |_p \right)$:

$$P_p |\Psi\rangle_{RK} = 0 \forall p \tag{97}$$

This implies that in $|\Psi\rangle_{RK}$, every state with a $|\Vert\rangle_p$ must have the same amplitude as the state having $| = \rangle_p$ - then they would cancel each other out on projection. If $|c\rangle$ is used to represent a valid covering of the lattice, and c' indexes over all states that differ by only local plaquette flips, then the RK ground state looks like:

$$|\Psi\rangle_{RK} = \sum_{c'} |c'\rangle \tag{98}$$

It seems that there are several states in the set indexed by c'. To classify them, we have to exploit the geometry of the torus that we described before.

Note 4.10.2.1 (Transition Graphs): Assume that we have two different, valid, dimer coverings of the lattice - $|c\rangle_A$ and $|c\rangle_B$. Let us color the dimers in the former with red, and the latter with blue (say), and then overlay them. Due to the covering rules, each site will now have two dimers of different colors connected to it, with the simplest being the case where the dimer is on the same bond in both coverings - we call this a loop of length 2. Otherwise, we will inevitably be able to draw a graph over alternating color bonds, which terminate into loops of various sizes. This is called a transition graph because if we "shift" the colors along all the graph lines, we will have transitioned between the coverings. The important part for our purposes is this - transition graph loops may grow, shrink, merge, or split under local plaquette flips. But, crucially, any transition graph loop that winds around either of the two directions on the torus can never be annihilated/closed using local plaquette flips. These require non-local moves, and thus any "flipping term" operator (which includes Eq. 95) can not take us between states that have different numbers of such "non-contractable" loops.

As a first step, we need to (arbitrarily) decide upon a specific plaquette. From this plaquette, we extend two perpendicular lines cutting across the bonds, and winding around the torus. We call these the x and y reference lines, based on our preexisting coordinates.

Now, we assign a direction to the dimers. Say, $A \to B$ is the "forward" direction. Now, we count the number of dimers being cut by the y reference line, going from left-to-right ($N^y_\rightarrow$) and right-to-left ($N^y_\leftarrow$). Let us define:

$$W_y = (N^y_\rightarrow - N^y_\leftarrow) \bmod 2 \tag{99}$$

Similarly, we define a quantity for the x reference line:

$$W_x = (N^x_\rightarrow - N^y_\leftarrow) \bmod 2 \tag{100}$$

We call these *topological fluxes*. Notice that (W_x, W_y) is a *topological invariant* under local plaquette flips:

- If a local flip occus in a plaquette that is not cut by a reference line, nothing changes.
- If a local flip occurs in a plaquette that is cut by a reference line, it will either take away a pair of opposite direction dimers or add a pair of opposite direction dimers. Since it can not just take away one dimer from the plaquette, (W_x, W_y) remains the same.

Thus, the total Hilbert space of this RVB system is split into mutually inaccessible "topological sectors", each labelled by (W_x, W_y) provided a well-defined set of reference lines - each of these sectors is massively degenerate. Since the topological fluxes can only take two values each, we have 4 sectors in total for the square lattice on a plane, with the previously prescribed PBC.

The RK Ground state is thus "liquid-like" RVB state, where only local plaquette flips dominate the dynamics, restricting any such state to its own topological sector.

It is currently conjectured that the RK point is only part of an extended spin liquid phase in 3 or more dimensions, or on frustrated 2D systems - i.e. it is only a boundary point between the long-range ordered phase (like staggered or columnar) and the spin liquid phase.

4.10.3. Fractionalized excitations in the RK model

In conventional local magnetic moment (i.e. spin) systems, we have spin-1 Nambu-Goldstone bosons as the low-energy excitations - magnons.

To study the excitations in the RK model, we first engineer a deliberate violation of the 3-in-1-out/1-in-3-out dimer covering rules by introducing *monomers*. It is represented by a site that has no dimers connected it to any of its neighbours. Thus, removing a dimer from a valid covering will result in the creation of two monomers in adjacent sites. These monomers may be thought of a spins that are in spin-triplet states, in our current context.

In an ordered phase, like the staggered phase, moving a monomer pair apart incurs an energy cost that grows linearly with the distance between them - which means that the pair of monomers are *confined* in this phase (See Section 4.2). This is due to such movement breaking up the ordered dimer covering, leaving behind unfavourably arranged dimers.

In the liquid phase, however, so such "favourably" arranged chain exists as it is an RVB state. This leads to the monomers becoming deconfined and they are the fundamental excitations in this phase. Note that each such monomer carries an effective spin-1/2, and is called a *spinon*. If we were to take a spin away from a site, leaving a hole, spin/charge excitation seperation (like a Luttinger liquid) may also be achieved.

4.10.4. Z_2 spin liquids

Dimer models, such as the RK model defined before, can give rise to emergent gauge fields (See Section 4.9) and just like classical gauge theories (example: Electromagnetism) the gauge field that describe the system have a redundancy in mathematical description. Gauge transformations, such as Eq. 28, do not change physical observables but do change the mathematical description of the system.

Similar to classical electromagnetism, such gauge decriptions are sought out because they have the potential (no pun intended) to make some otherwise complicated complicated calulations simple.

In gauge descriptions of quantum matter, we begin by extending the Hilbert space and describing the system in terms of a new set of operators on the extended space. This space offers an overcomplete representation of the physical system - multiple "names" are ascribed to the same state. A gauge transformation, then, is only a transformation that maps a "name" of a physical state to another "name" for the same physical state.

In this section, we will attempt to use a Z_2 Gauge Theory to find excitations in a quantum dimer model that would be difficult to show using other methods. A motivation for the Z_2 theory would not be provided here, as that is beyond the scope of the report. Only a description of the theory, and how it applies to the Dimer model will be discussed.

On our bipartite lattice, we introduce a variable s_{ij} that is assigned to each bond between sites i and j. We have the condition:

$$s_{ij} = s_{ji} = \pm 1 \tag{101}$$

Think of the value of s_{ij} being -1 if that bond has a dimer, and $+1$ otherwise. So, an assignment of values for s_{ij} over the entire lattice defines a physical state of the system.

Now, we define a *gauge transformation* as:

$$\widetilde{s_{ij}} = W_i s_{ij} W_j^{-1} \tag{102}$$

Where $W_i = \pm 1 \forall i$. Since we have two possible values of this "transformer" W_i, call this a Z_2 gauge theory.

So, an assignment of values for s_{ij} over the entire lattice defines a physical state of the system as is typical in a Quantum Dimer Model. What we are claiming here is that if the transformation Eq. 102 were to be applied to the entire lattice we would have a new set of $\tilde{s}_{ij}$ that would be a different "name" or label for the exact same physical state.

Now, we must estimate the size of the extended Hilbert Space defined allowing for the Z_2 gauge theory. We work with PBC, as before.

Note that:

- Each plaquette on the square lattice has 4 sites, which means that there are 4 possible labels i for W_i transformations. Since W_i can have only two different values, we haave a total of $2^4 = 16$ available gauge transformations.
- Each plaquette also has 4 bonds, each of which can be assigned a value of s_{ij}. Since s_{ij} is allowed to have only two values, we have a total of $2^4 = 16$ physical states.

Since the number of states is equal to the total number of gauge transformations possible, it must imply that there is only *one* physical state possible for the plaquette - all of the 16 states would be gauge-equivalent via the 16 possible transformations. But this is an overcounting - the transformations where $\mathrm{sgn}(W_i) = \mathrm{sgn}(W_j)$ in Eq. 102 map between identical gauge (not physical) representations. This is an indentity operation, and we may allow only one such identity gauge transformation, not two - this is done to give the set of Group transformations a group character. Also, the trasformations where the signs of the two W_i factors are flipped is the same transformation (if the other two factors are unchanged). This is also an overcounting, and is to be disallowed.

Thus, we end up with a reduction in the number of gauge-equivalent states by a factor of 2. There are now 8 such states, and thus $16/8 = 2$ physical states on each plaquette. Extending to a lattice with N sites, we note that there are $2N$ bonds. Reducing the total number of transformations by a factor of 2 as before, we may deduce that there are:

$$\frac{2^{2N}}{2^N/2} = 2^{N+1} \tag{103}$$

physical states on this lattice.

We may now define a quantity called a Z_2 *flux* on each p-th plaquette:

$$F_p = \prod_{\substack{i,j \in p \\ i \neq j}} s_{ij} \in \{-1, +1\} \tag{104}$$

Which is simply the product of all the values on the 4 bonds of the plaquette. Note that the value of this flux for two adjacent plaquettes, denoted by $F_{p+p'}$ is given as:

$$F_{p+p'} = \prod_{\substack{i,j \in P_{p+p'} \\ i \neq j}} s_{ij} \tag{105}$$

Where $P_{p+p'}$ is a set that contains the indices of all the sites on the *rectangle* formed by joining the two plaquettes - the shared bond between adjacent plaquettes is ignored since it would show up twice and $(\pm 1)^2 = 1$. If we calculated the net Z_2 flux for the lattice, every one of the $2N$ bonds would show up twice. Therefore:

$$F_{total} = \sum_p F_p = 1 \tag{106}$$

Since there are the same number of plaquettes as there are lattice sites. This constraint on the total Z_2 flux tells us that we are actually free to choose $N-1$ fluxes - not N as previously assumed. Therefore, there are 2^{N-1} configurations of fluxes.

Since a value of flux effectively labels a plaquette's physical state here, we have a total of $2^{N+1}/2^{N-1} = 4$ distinct physical states that are possible for this system. This factor of 4 is related to the genus of the surface on which the lattice lives - it has to do with the two distinct directions in which topologically non-trivial (non-contractable) loops can be drawn on the torus.

For each fixed pattern of fluxes on the lattice, there are 4 possible configurations of s_{ij} values on the lattice, as $2^{N-1} \cdot 4 = 2^{N+1}$, which is the total number of configurations possible for the unconstrained lattice as previously shown.

Let us take a set of values s^0_{ij}. Given x and y reference lines on the latticce, we define X_{ij} as follows:

$$X_{ij} = \begin{cases} -1, \text{The bond ij cuts x-reference line} \\ +1, \text{Otherwise} \end{cases} \tag{107}$$

Defining Y_{ij} similarly using the y-reference line.

Now, the 4 distinct configurations of s_{ij} for each flux configuration are given by:

$$s^{(W_x,W_y)}_{ij} = X^{W_x}_{ij} Y^{W_y}_{ij} s^0_{ij} \tag{108}$$

Where $W_x, W_y \in \{-1, +1\}$ and they correspond to the topological fluxes defined in Eq. 100 and Eq. 99. In terms of these "winding numbers", one can view going between the physical states as:

- Inserting a flux through the inside of the torus.
- Inserting a flux through the hole of the torus.

Surfaces with higher genus g have a higher topological degeneracy owing to $2g$ unique topologically distinct loops that can be drawn onto the surface, leading to a degeneracy of 2^{2g}.

Now, we need to write down the effective low-energy Hamiltonian for the system that we have described through the Hilbert Space. To do this, we first look for the lowest-order gauge invariant terms. F_p are defined in Eq. 104 is gauge invariant, and we define an operator σ^x_{ij} acting on s_{ij} to flip its sign:

$$\sigma^x_{ij} s_{ij} = -s_{ij} \tag{109}$$

Which is also gauge invariant - Notice that any gauge transformation that is not identity will flip the sign of the operator σ^x_{ij} as well as the value of s_{ij} that it is acting on, thus preserving the action of the σ^x_{ij} operator. This leads us to a low-energy Hamiltonian:

$$H_{Z_2} = -\nu \sum_p F_p - t \sum_{<ij>} \sigma^x_{ij} \tag{110}$$

In making this Hamiltonian, we have simply sought to include the lowest-order terms with arbitrary energy scale parameters (ν, t).

In the $\nu \gg |t|$ case, the ground state will have $F_p = 1 \forall p$. In this case, the lowest energy excitations would involve changing the signs of a pair of fluxes (Note that due to the $F_{total} = 1$ constraint this is required for us to change out from the ground state while satisfying the constraint). This can be done by acting with σ^x_{ij} on some bond, which will change the fluxes of the adjacent plaquettes that share that bond by opposite signs. These excitations are known as Z_2 vortices, or *visons*. The second t-term has the effect of causing these visons to hop. The σ^x_{ij} operators commute with each other, giving these visons bosonic exchange statistics. These are the new excitations that we were seeking, and taking the view of fluxes defined on plaquette bonds would not be obvious/easy to show without the Z_2 gauge theory approach.

Winding numbers do not appear naturally in Eq. 110, and so it would apear that the energy of the system is indifferent to (W_x, W_y) and we have fourfold degeneracy on the torus. This is correct in the thermodynamic limit of $N \to \infty$, but it is incorrect for finite sized lattices. Note that in a $L \times L$ finite lattice, we can go between the 4 states linked via Eq. 108 by applying the σ^x_{ij} operator to all the bonds s_{ij} that cross the x- or y-reference lines. In effect, this would be equivalent to creating a pair of visons, taking them around the torus (in a direction determined by our choice of which reference line we wish to do this to), and then annihilating them. Perturbation theory predicts that this will lift the fourfold degeneracy by $\sim e^{-cL}$ for some constant c.

Visons, just like the monomers in the RK model, can not be created individually by any local operator - they must be created in pairs, with opposite winding numbers assigned to them. This is because a local operator can not change the winding numbers of the system, and is clear from the action of σ^x_{ij} on a bond described before.

Let us make a pair of monomers by splitting a dimer in two. We also create a pair of visons. Now, leaving one of the monomers stationary, we take the other and move it *around* the vison, and then take it back to be annihilated by the other monomer to form a dimer again. This action would be equivalent to flipping all the dimers along the path taken by the moving monomer. Due to this path being "around" one of the visons, it must cut the path connecting the two visons an odd number of times. This has the effect of generating a "geometric phase" of π (or, a sign change) in the wavefunction that describes the system. Due to this effect, monomers and visons are said to have *mutual semionic statistics*. Semions are a type of anyon.

Note that in the abscence of any visons, the monomers have bosonic exchange statistics. This seems contrary to the spin-statistics theorem, which requires spin-1/2 particles like monomers to have fermionic statistics. But the spin-statistics theorem requires the theory to be (pseudo) Lorentz invariant - something that is broken explicitly in this model. As the last paragraph describes, the bound state of a monomer and a vison has *fermionic* exchange statistics, which is non-intuituve granted that both of them have *bosonic* exchange statistics in the absense of each other.

In the opposite regime of $t \gg |\nu|$, the system tries to maximize the "flippability" of the plaquettes, and the model becomes equivalent to a a lattice of uncoupled spins that have a magnetic field in the x-direction. This results in each bond being in a quantum superposition of $+1$ and -1 values, and the ground state being a sum of all such configurations. This phase is topologically trivial, and has none of the Z_2 gauge theory character in the low-energy physics - this is called the *confined* phase.

Kitaev's toric code is a model for a system in a *deconfined* phase (i.e. a $\nu \gg |t|$ model), which makes any system obeying the model a valid candidate as a *qubit* for quantum computation. The topological order stabilizes the system naturally against local noise - localizing the system in one of the 4 physical states. In the thermodynamic limit, such a system would act as a perfectly noise-resistant qubit.

4.11. $U(1)$ spin liquids

Let us consider a bipartite, 3-dimensional lattice with coordination number z. There are two sublattices, A and B as usual, and all bonds on the lattice are between sites that belong to different sublattices.

Let us define the "positive direction" to be from a site in the A sublattice, to a site in the B sublattice. We intend to create a Quantum Dimer Model on this latticce, so the fundamental degree of freedom is on the links. We utilize $n_{ij} = \frac{1-s_{ij}}{2}$ as the occupation number of a bond - it is 1 if there is a dimer on the bond and 0 otherwise.

To enforce the dimer covering properly, we require:

$$\sum_{j \in NN(i)} n_{ij} = 1 \forall i \in A \tag{111}$$

and,

$$\sum_{j \in NN(i)} n_{ij} = 1 \forall i \in B \tag{112}$$

i.e. only one dimer touches each site. $NN(r)$ denotes the set of nearest neighbour lattice site of r.

This system can be described using an emergent gauge theory, quite like electromagnetism that is discretized. We first define the "electric field" on a bond:

$$E_{ij} = n_{ij} - \frac{1}{z} \tag{113}$$

The reason that we are subtracting $\frac{1}{z}$ here is as follows: We require a quantity that we can construct out the dimer occupation numbers that give us a measure of an electric field that can point "in" or "out". Since our occupation numbers are themselves non-negative, this can be done only by subtracting out the background "field". In the case where all the lattice points are uncorrelated, each bond has a $\frac{1}{z}$ probability of hosting a dimer. We treat this uncorrelated situation as the background field and subtract it from the occupation numbers. This leaves us with:

$$E_{ij} \in \left\{1 - \frac{1}{z}, -\frac{1}{z}\right\} \tag{114}$$

For a cuboc lattice with $z = 6$, the electric field can only take the values $\frac{5}{6}$ or $-\frac{1}{6}$.

Now, we can define the divergence of the Electric field at a point on the lattice. In the discrete case, this simply means that we need to add all the fields coming in/going out of the site.

$$\left(\nabla \cdot \hat{\boldsymbol{E}}\right)_r = \sum_{r' \in NN(r)} \varepsilon_{r,r'} E_{rr'} \tag{115}$$

Where $\varepsilon_{r,r'}$ is $+1$ if the link r, r' is oriented “away from” the site r, and -1 if the link r, r' is oriented “toward” the site r. For $r \in A$, all links point outwards, and the opposite is true for $r \in B$. Combining this with the covering constraint and the allowed values of the Electric fields, we can show that:

$$\left(\nabla \cdot \hat{\boldsymbol{E}}\right)_i = 0 \forall i \in A, B \tag{116}$$

Comparing directly with Gauss' Law, we can see that there are no “electric charges” in the ground state of this system.

To be able to create “charges” (i.e. excitations) in this system, we require a *conjugate momentum* of the electric field that we have defined. We introduce a vector potential A_{ij} to serve this purpose. Thus, we impose the commutation relation:

$$\left[A_{ij}, E_{kl}\right] = i\delta_{ij,kl} \tag{117}$$

Due to E_{ij} being allowed to take only discretely spaced values, the conjugate variable A_{ij} is prohibited from taking continuous values that extend to infinity. It must be a *compact phase variable*, restricted to take values in $[0, 2\pi)$. To have gauge invariance, we claim that $A_{ij} = A_{ij} + 2n\pi, n \in \mathbb{Z}$.

Now, we can use the BCH lemma to show that:

$$\begin{aligned} e^{iA_{ij}} E_{ij} e^{i-A_{ij}} &= E_{ij} - 1 \\ e^{-iA_{ij}} E_{ij} e^{iA_{ij}} &= E_{ij} + 1 \end{aligned} \tag{118}$$

The operator $e^{iA_{ij}}$ adds a unit flux, i.e. places a dimer on the link $i \to j$. $e^{iA_{ij}}$ does the opposite and removes a dimer from the bond. SImilarly, $e^{-i\theta E_{ij}} A_{ij} e^{i\theta E_{ij}} = A_{ij} + \theta$.

In normal electrodynamics, we define the magnetic field as $\boldsymbol{B} = \nabla \times \boldsymbol{A}$. Inspired by this definition, we can define a similar magnetic field in our model. However, in the discrete case the curl becomes the *circulation* of the vector potential A_{ij} on an elementary closed loop - which is still a 2×2 plaquette as before. Since we are calculating the circulation of a *vector* field (which A_{ij} do satisfy, if we account for the $A \to B$ directedness), we can, and must, define the field as:

$$B_p = \sum_{i \in P_p} \eta_{p,i} A_i \tag{119}$$

Here p is the plaquette index, and P_p denotes the set of links/bonds on the plaquette. We take a counter-clockwise direction to ccalculate this circulation, and $\eta_{p,i}$ enforces this - it is $+1$ if the counter-clockwise direction is also a $A \to B$ direction, and it is -1 otherwise.

Now that we hae established the appropriate variables of the gauge theory, we formulate the gauge transformation for this theory. Defining an arbitrary scalar function over the lattice points Λ_i, we have a unitary operator for gauge transformations as:

$$U(\Lambda) = \exp\left[i \sum_j \Lambda_j (\nabla \cdot E)_j\right] \tag{120}$$

Defining q_i to be the electric charge at the i-th site, we have have the action of this unitary on wavefunctions defined for the system.

Now, we take the lowest terms consistent with gauge invariance, and form a Hamiltonian:

$$H = -K \sum_p \cos(B_p) + K' \sum_i q_i^2 + U \sum_{ij} E_{ij}^2 \tag{121}$$

Where the energy scale parameters (K, K', U) define the problem. The $\cos(B_p)$ term appears as we need to respect the compactness of A_{ij} - B_p can not appear directly, but putting it inside a cosine function makes the contribution compact due to the nature of the cosine function.

In a $2 + 1$ dimensions (i.e. 2 spatial and 1 time dimension) theories, the excitations are always confined. This serves as an explanation for why there is no sqin liquid phase in dimer models in 2D bipartite lattices. In $3 + 1$ dimensions, this model can have a deconfined phase for a small range of values in U. Taking the continuum limit with no charges, it is even possible to recover the form of the typical electromagnetic hamiltonian:

$$H \approx \int \mathrm{d}^3x \left[\frac{\varepsilon^*}{2}|\boldsymbol{E}|^2 + \frac{1}{2\mu^*}|\boldsymbol{B}|^2\right] \tag{122}$$

The values of permeability and permittivity have no reason to match the values in vacuum, but the form of the Hamiltonian does seem to suggest the existence of a photon-like excitation with the typical linear dispersion, and two transverse polarizations. This also tells us that the deconfined phase has no topological order, as typically topologically ordered phases have a gap between the degenerate ground states and the excitated states. It is also possible to show that both the magnetic (monopoles exist here) and electric charges both interact among themselves via Coulomb-like potentials - thereby gaining the phase the name "Coulomb Phase".

Different routes exist to establishing a theory for quantum spin liquids, such as the Kitaev honeycomb model, which tries to model the effects of strong spin-orbit coupling. A thorough discussion of this model is beyond the scope of this report.

4.12. The Fractional Quantum Hall Effect (FQHE)

As we discussed in Section 4.4, 2DEGs require a small amount of disorder for the IQHE to be observable. As technology progressed, it soon became possible to have 2DEGs that were "cleaner". Simultaneously, stronger fields became availble in the laboratory.

Under these conditions, *additional plateaus were observed* at *fractional filling fractions* $v = \frac{p}{q}$. This is called the Fractional Quantum Hall Effect. Most of the fractions have an odd denominator, but there are some exceptions to this - the most notable being a prominent plateau at $\nu = 5/2$.

FQHE was the first experimental context in which fractionalization and topological order were discovered, and it is an important example of these phenomena in a system that is more than just a single dimension.

In the IQHE, the many-body wavefunctions that describe the system at particular integer values of ν are simply antisymmetrized products of the single-particle wavefunctions - the Landau levels. To explain FQHE, Robert Laughlin proposed a generalization of these many-body wavefunctions to describe the composite fermions formed from electrons bound to vortices that are threaded by a fixed amount of flux quanta. Using this, Laughlin was able to explain $\nu = \frac{1}{2p+1}, p \in \mathbb{Z}$ plateaus. For example, in terms of Laughlin's description, the $\nu = 1$ state was comprised of composite fermions that were formed of an electron and $2p$ flux quanta.

Moreover, since these composite fermions are bound states of a charge and a flux, we expect the exchange of these to incur an AB phase - implying that Laughlin states are Abelian anyons. Counting arguments can also show that certain states have quasiparticles with fractional charges. Any further discussion of the FQHE is well beyond the scope of this report.

5. Conclusion

[1] takes us through a wide ranging survey of the myriad phenomena that exist condensed states of matter, and the many techniques and methodolgies used to understand them theoretically.

The main takeaway from the extended discussion in the article is this: Almost any kind of particle, or field, can emerge as a collective excitation in a system that has sufficient correlation. This includes particles that have very unusual properties, such as fractional charges or spins, as well as particles that are not found in the Standard Model of particle physics. It requires significant insight, and new phenomenology to be able to describe increasing complex and exotic states of matter and elucidate the primary excitations that give them their character.

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